

case2_v13.0

August 21, 2025

1 Validation du model turbulent poreux en TrioCFD - Cas 1

1.1 Introduction

Validation made by : ALi Fadlallah

Report generated 21/08/2025

1.1.1 TRUST parameters

- Version TRUST: 1.9.5
- Binary used: /volatile/catC/af279222/TrioCFD-1.9.5/TrioCFD_opt (built in directory /volatile/catC/af279222/TrioCFD-1.9.5/tests/Reference/Turbulence/perso/case2/v13.0/build)

1.1.2 Description:

Rub reference case and the test cases

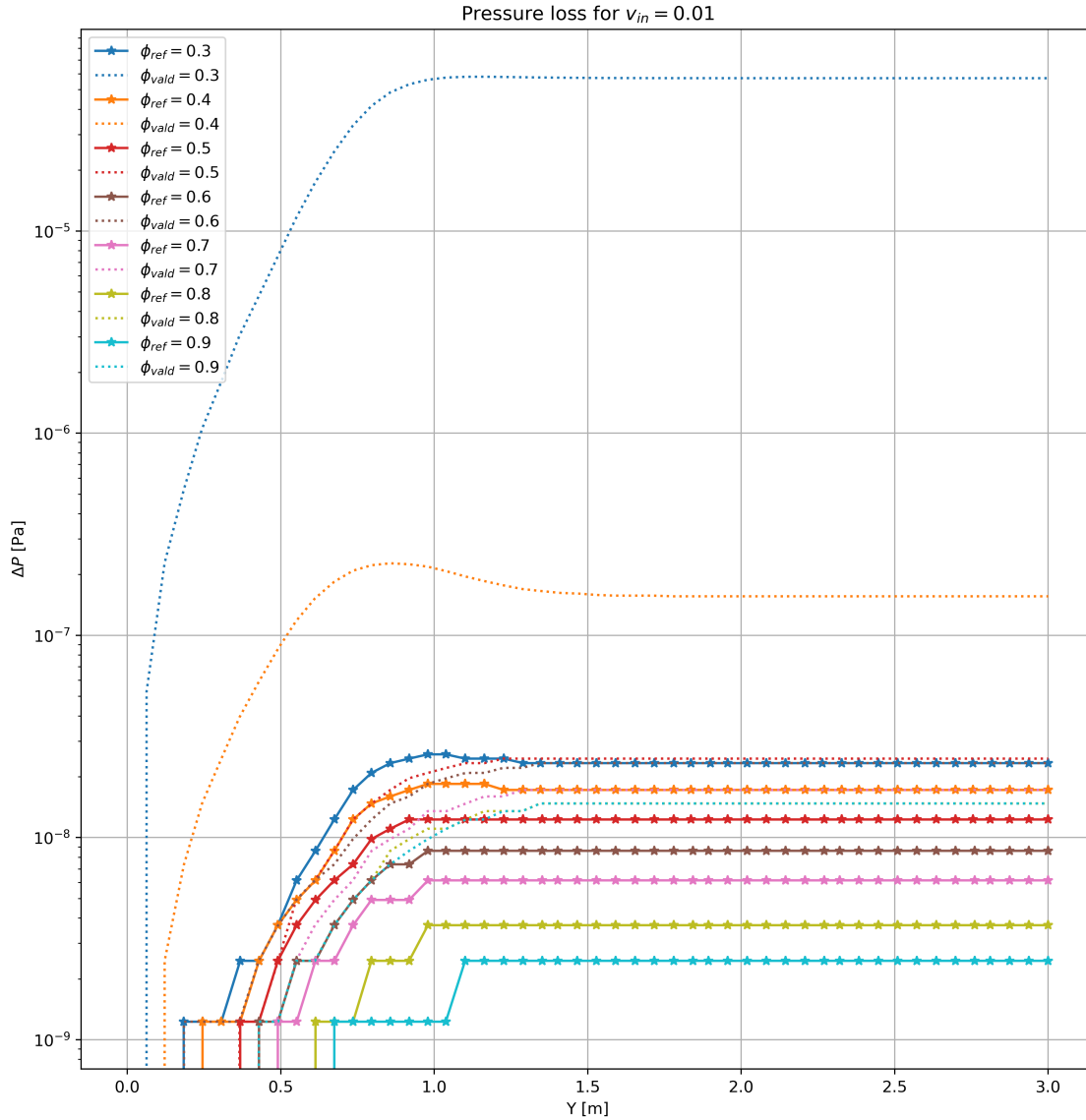
```
'\nrun.reset()\nrun.defaultSuite_ = run.TRUSTSuite(runPrepare=False)\n\nfor v_in in velocities:\n    for phi in porosities:\n        geometry = compute_geometry(phi)\n        geometry.update({ "v_in": v_in,\n            "Nx1": Nx1, "Nx2": Nx2, "Nx3": Nx3,\n            "Ny": Ny,\n            "Ny_PE": Ny_PE, "channel_width": channel_width,\n            "channel_height": channel_height,})\n        \n        run.addCaseFromTemplate("jdd_ref.data",\n            f"reference/vin_{v_in:.2g}/phi_{phi:.1g}",\n            geometry)\n\n\nrun.printCases()\n#run.runCases()\n#run.tablePerf()'
```

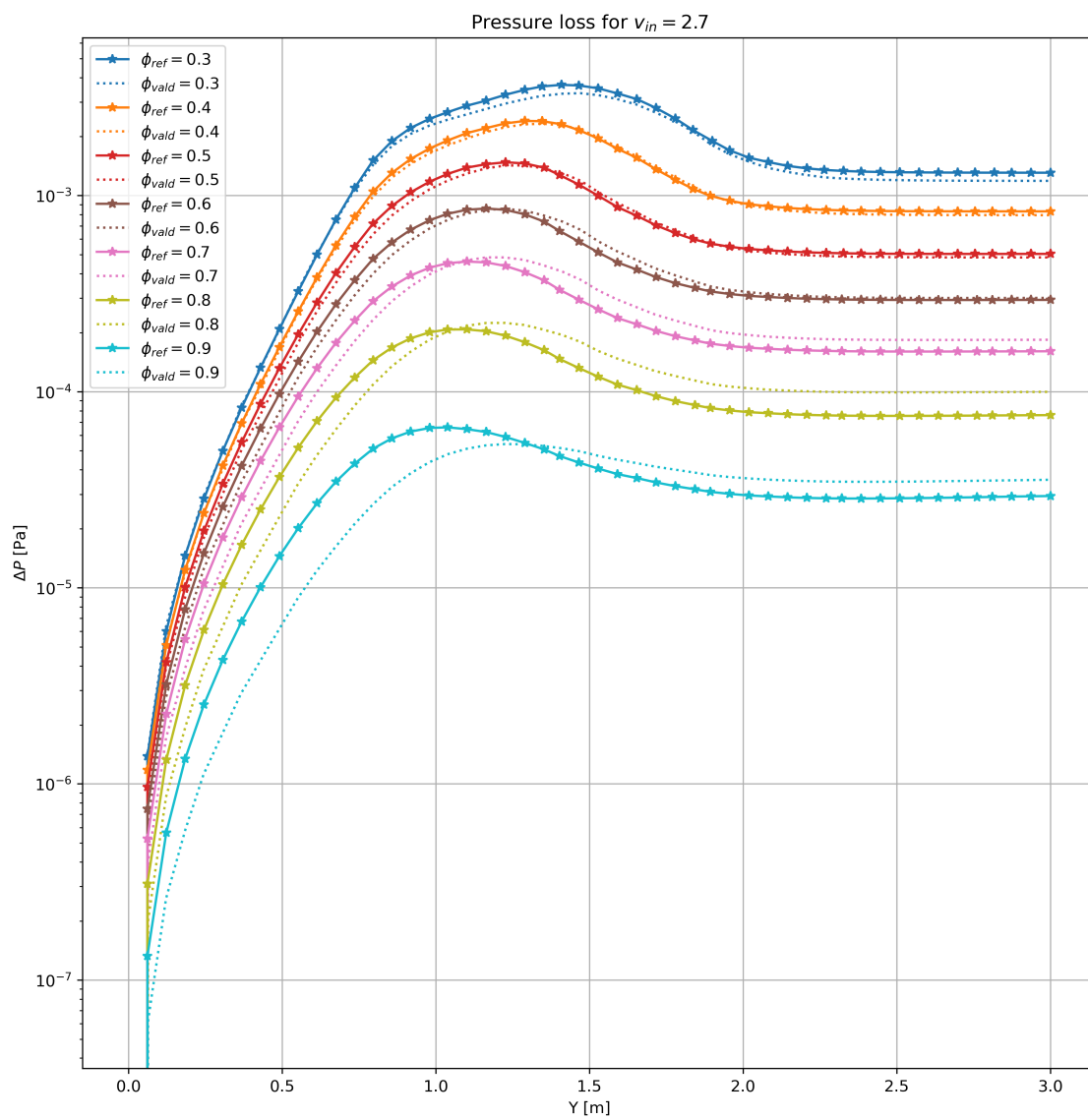
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'\n# Loop over all combinations\nfor c0, c1, v_in, phi in\n    itertools.product(c0_values, c1_values, velocities, porosities):\n        #case_name\n        = f"c0_{c0:.1g}_c1_{c1:.1g}/phi_{phi:.1f}"\n        \n        geometry = compute_geometry(phi)\n        run.addCaseFromTemplate(\n            "jdd.data",\n            f"validation/vin_{v_in:.2g}/c0_{c0:.1g}_c1_{c1:.1g}/phi_{phi:.1g}",\n            {\n                # Frottement poreux\n                "longueur": geometry["Lx2"],\n                \n                "c_0": c0,\n                "c_1": c1,\n                "v_in": v_in,\n                \n                "Nx": Nx,\n                "Ny": Ny,\n                \n                "Nx_PE_bas": Nx,\n                "Ny_PE_bas": Ny,\n                \n                "Nx_PE": Nx,\n                "Ny_PE": 2*Ny,\n                \n                "Nx_PE_haut": Nx,\n                "Ny_PE_haut": Ny,\n                \n                "porosity": phi,\n                "channel_width": channel_width,\n                "channel_height": channel_height,\n                \n                #
```

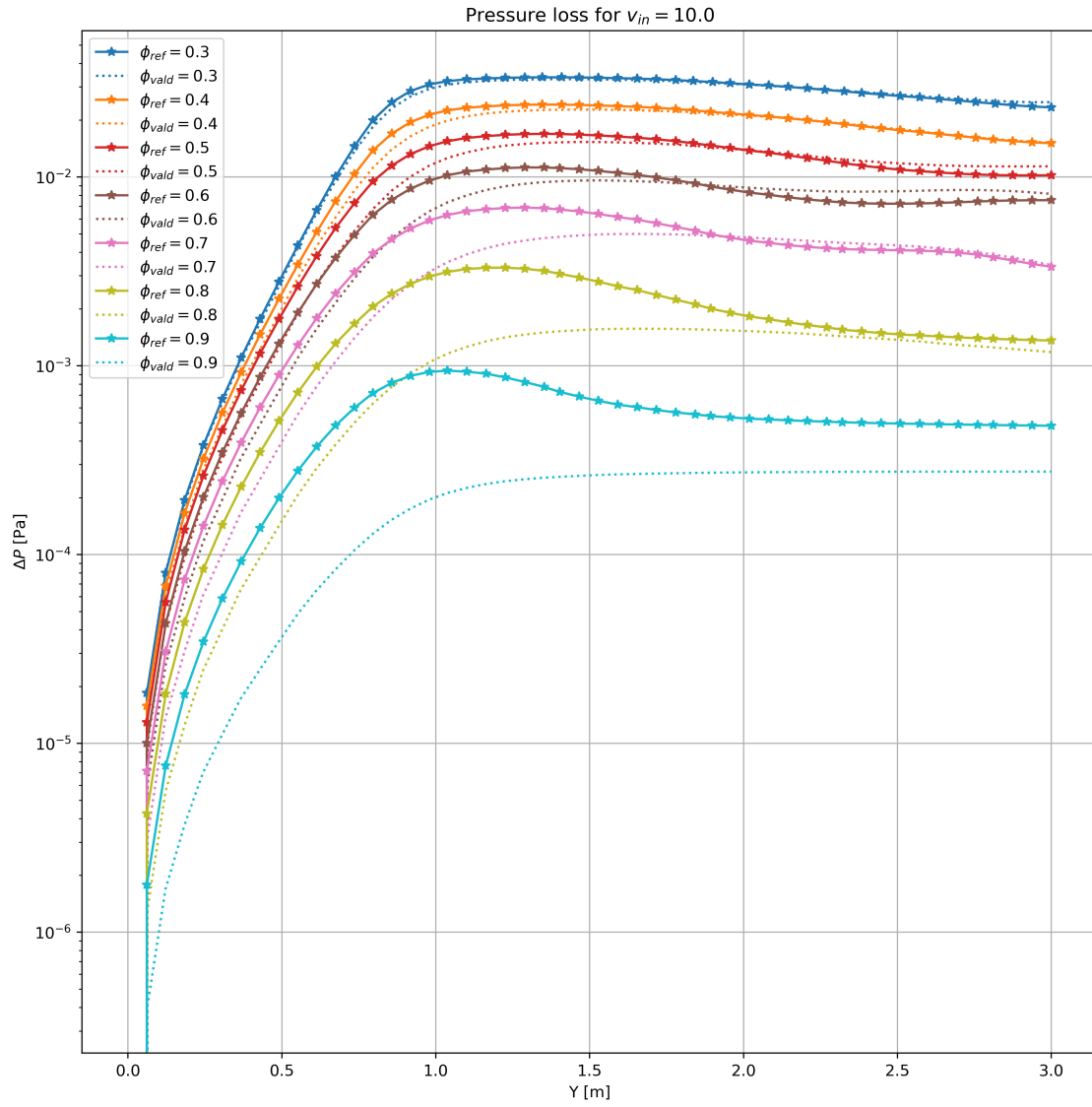
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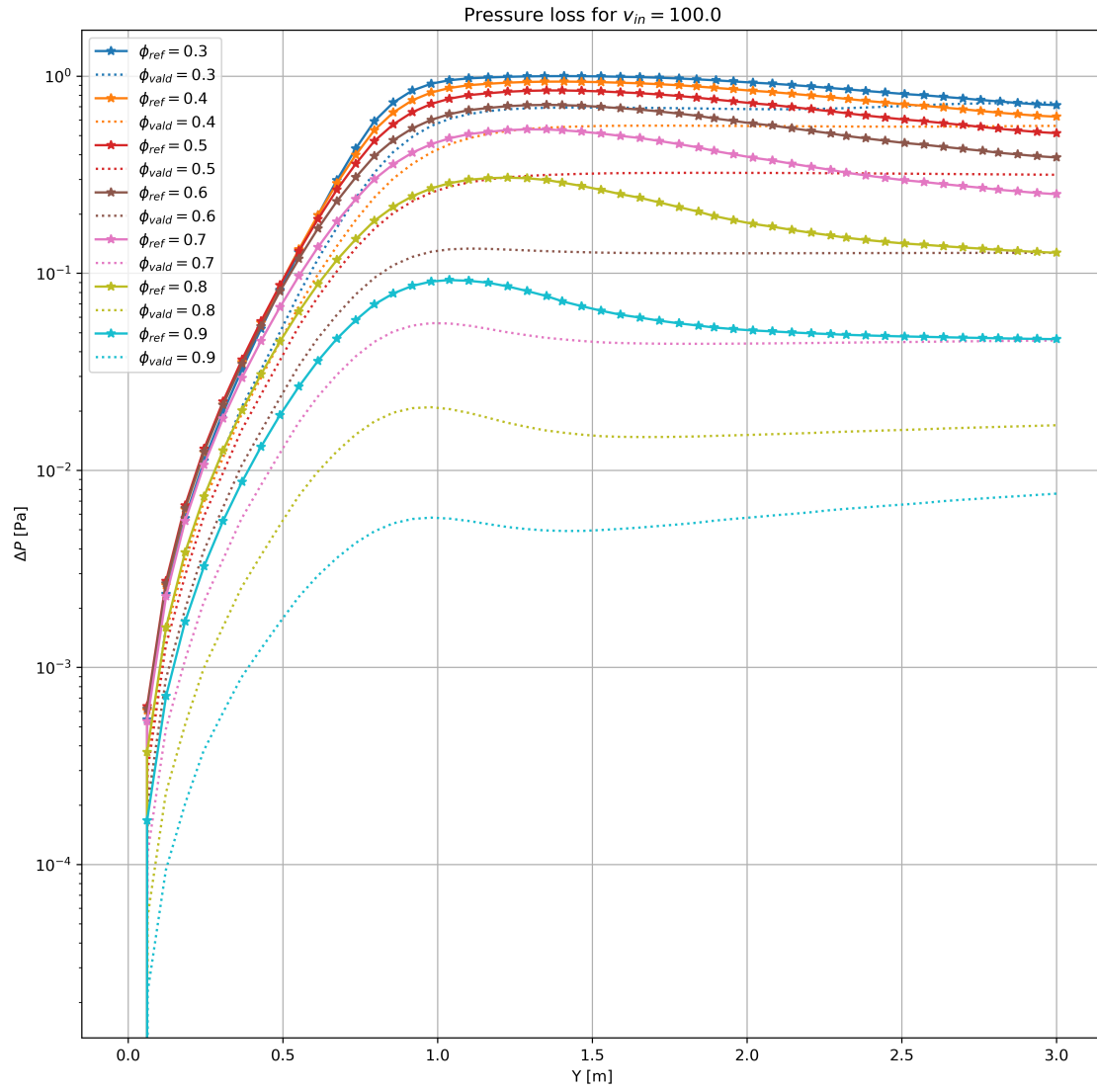
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"Y1": geometry["Y1"],\n
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# Sizes\n
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"Lx3": geometry["Lx3"],\n
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"Ly_PE_haut": round(geometry["Ly_PE"]/3.0 , 6),\n
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"solid":round(1-phi,6),\n
}\n
)\n\nrun.printCases()\nrun.runCases()\nrun.tablePerf()'

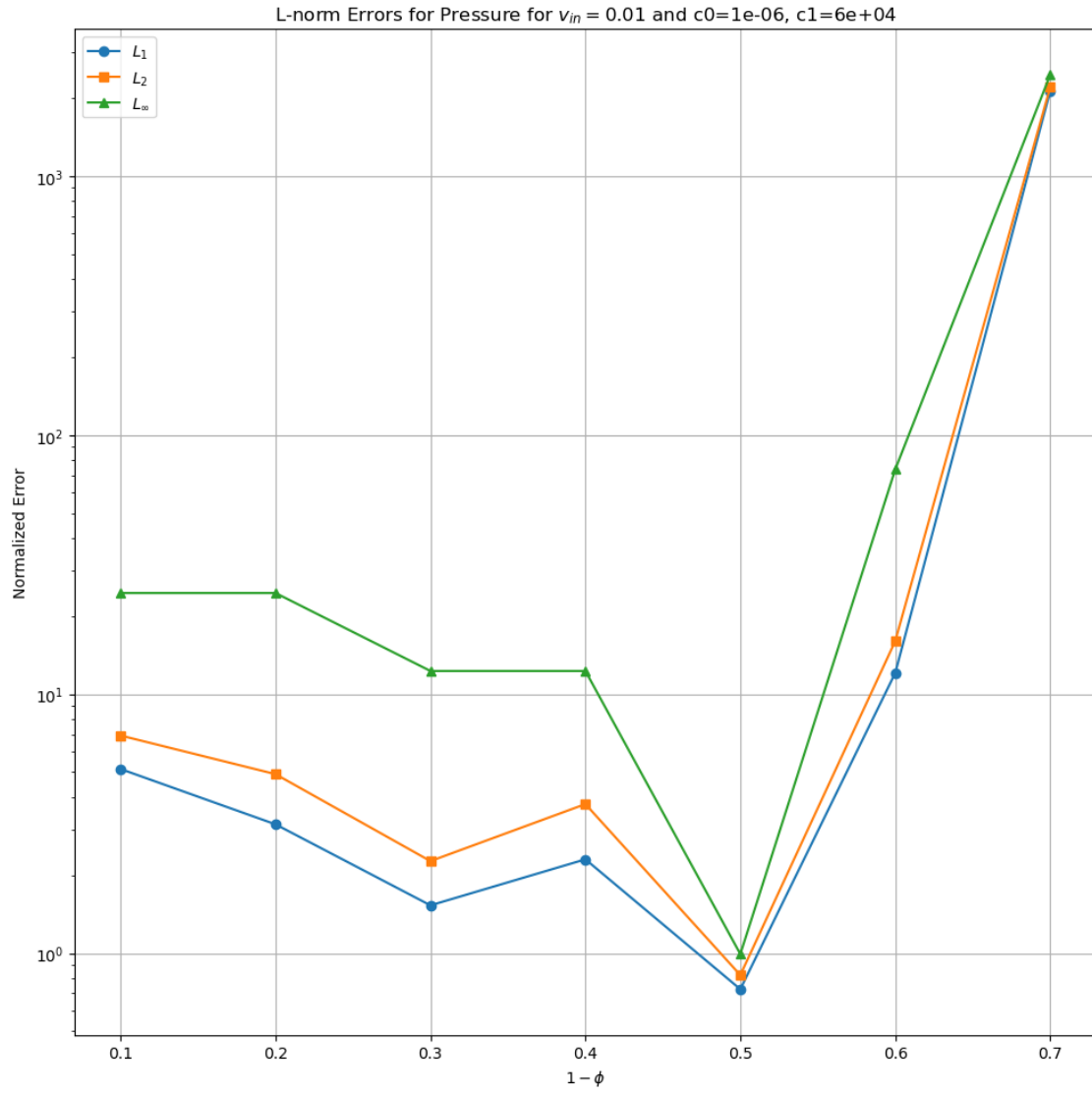
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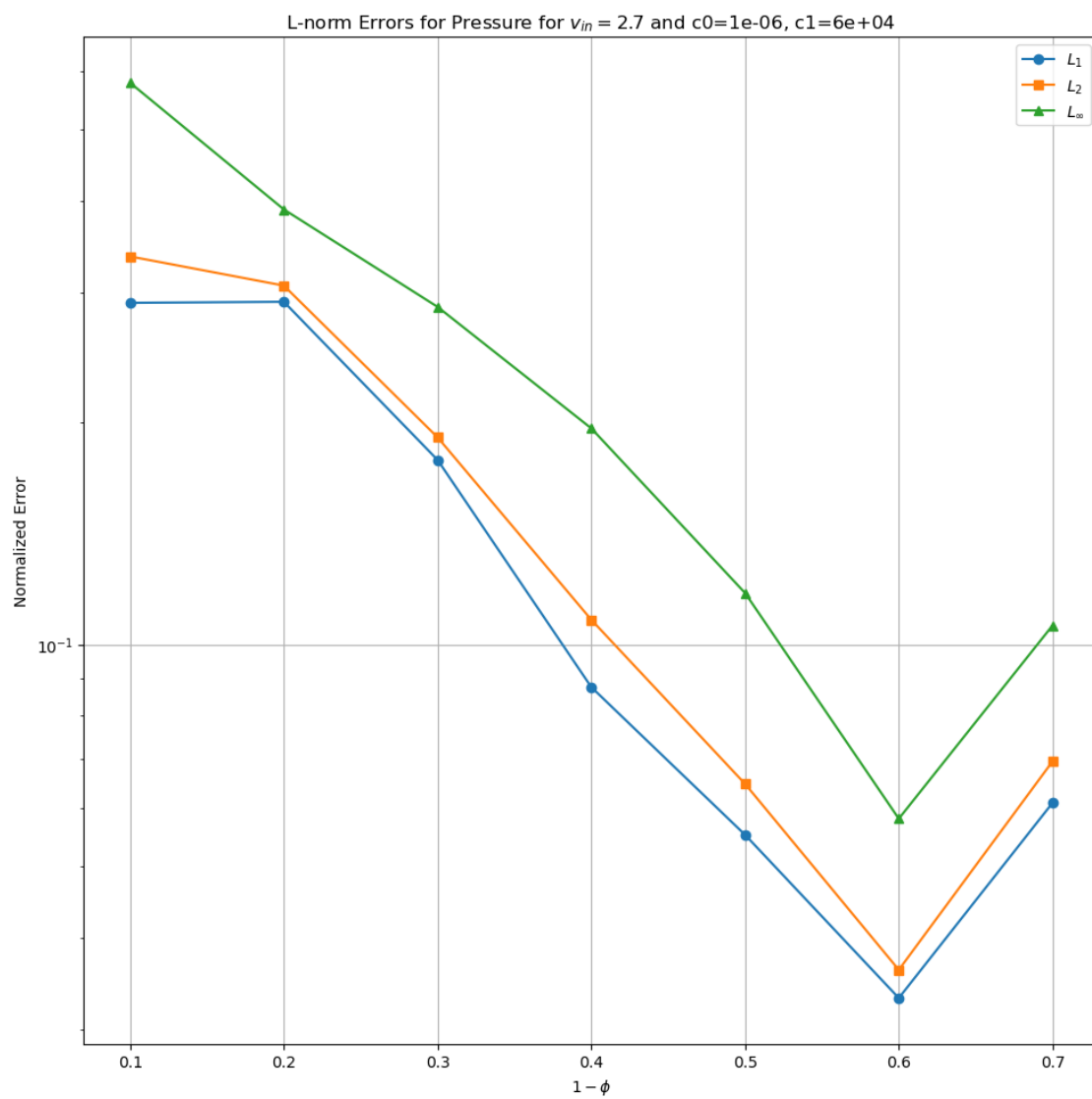


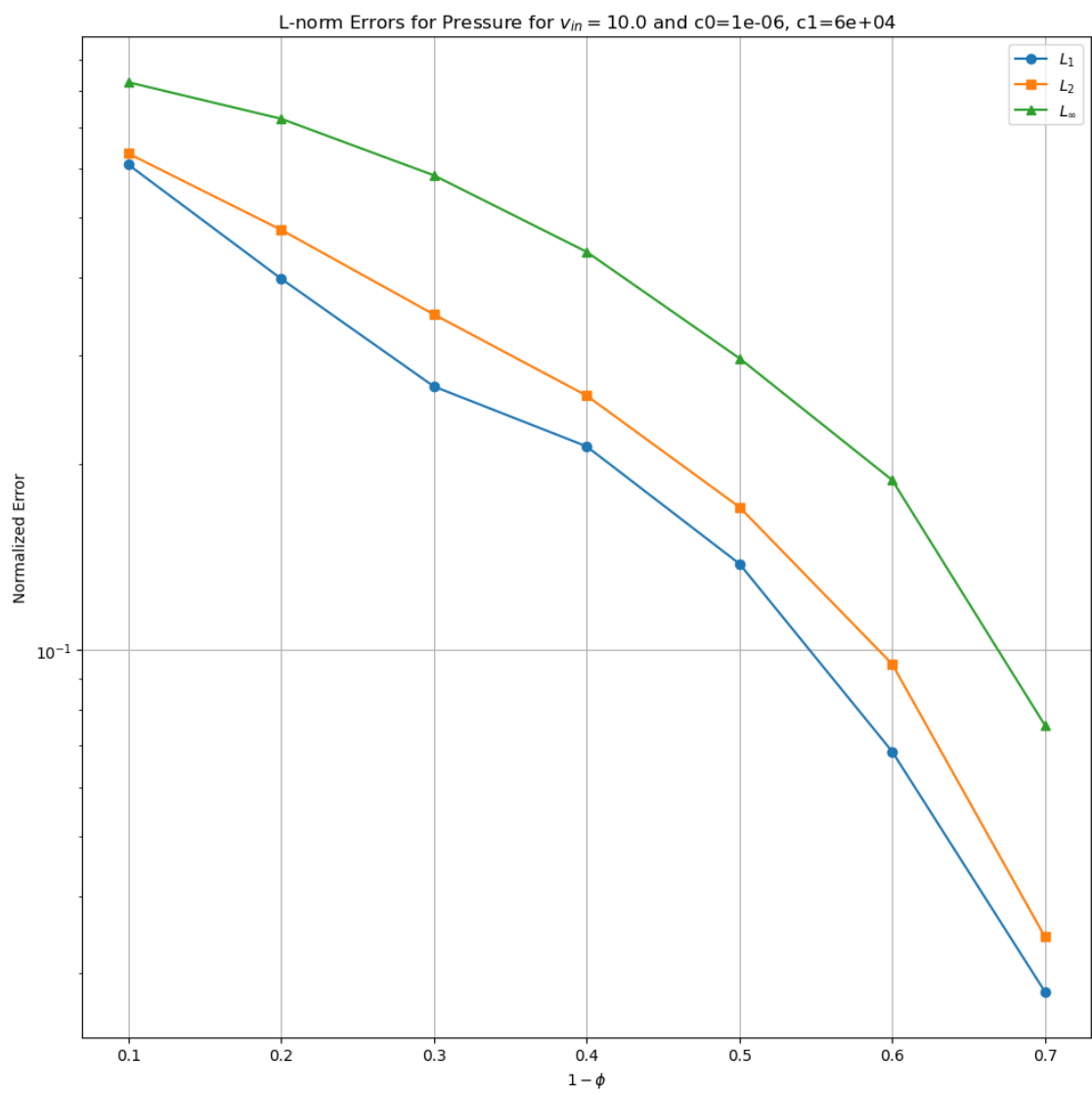


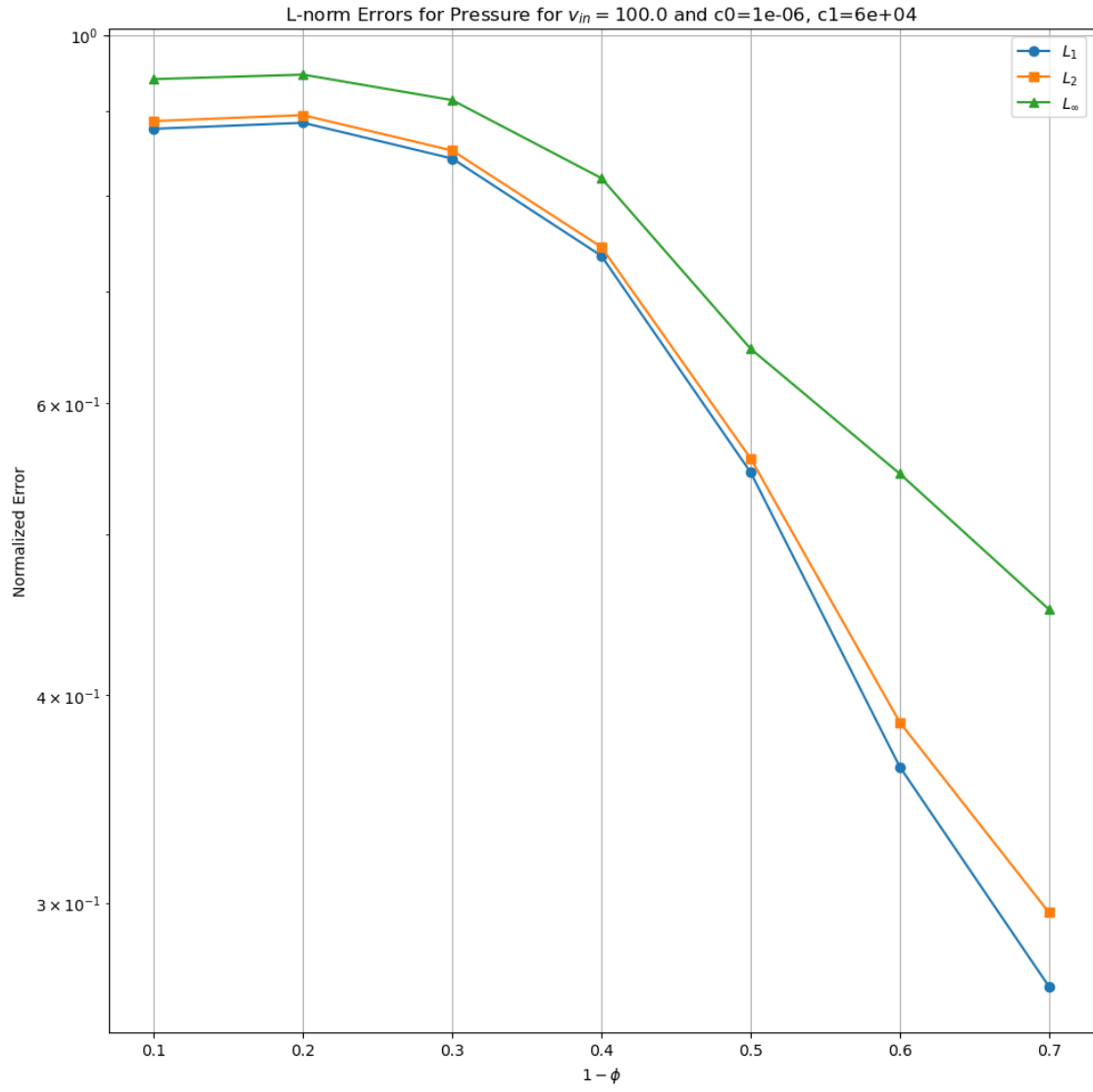


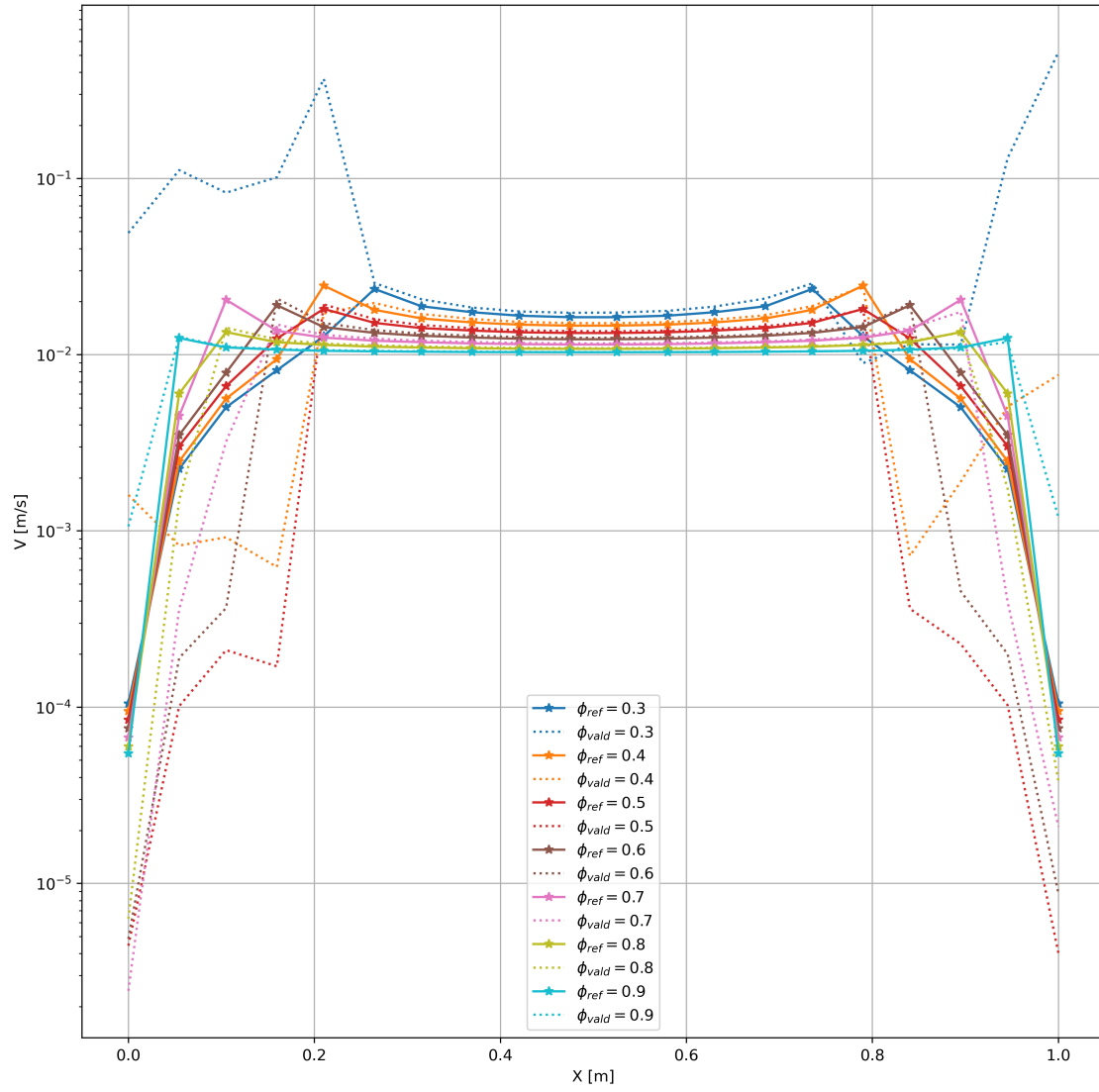


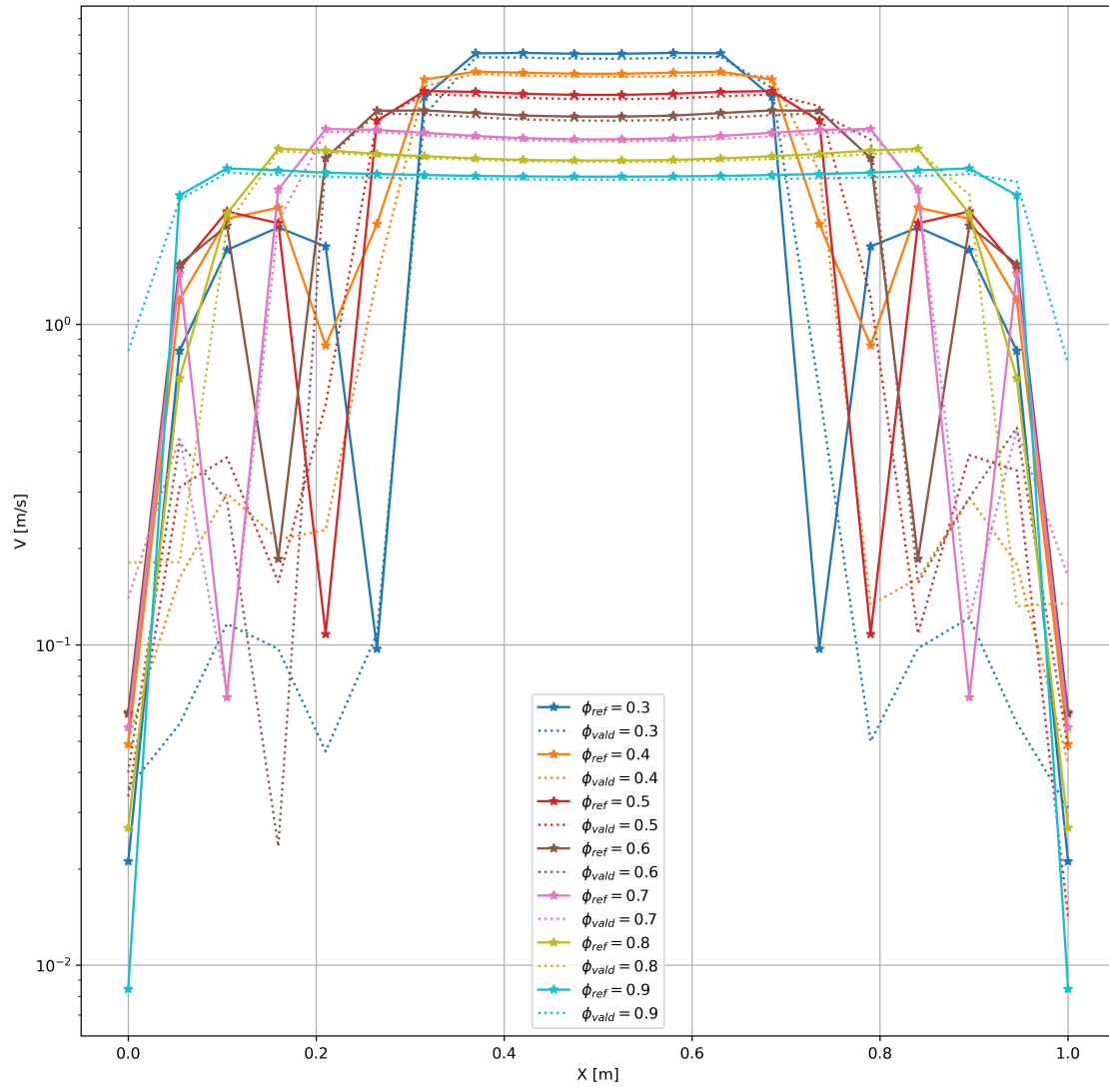


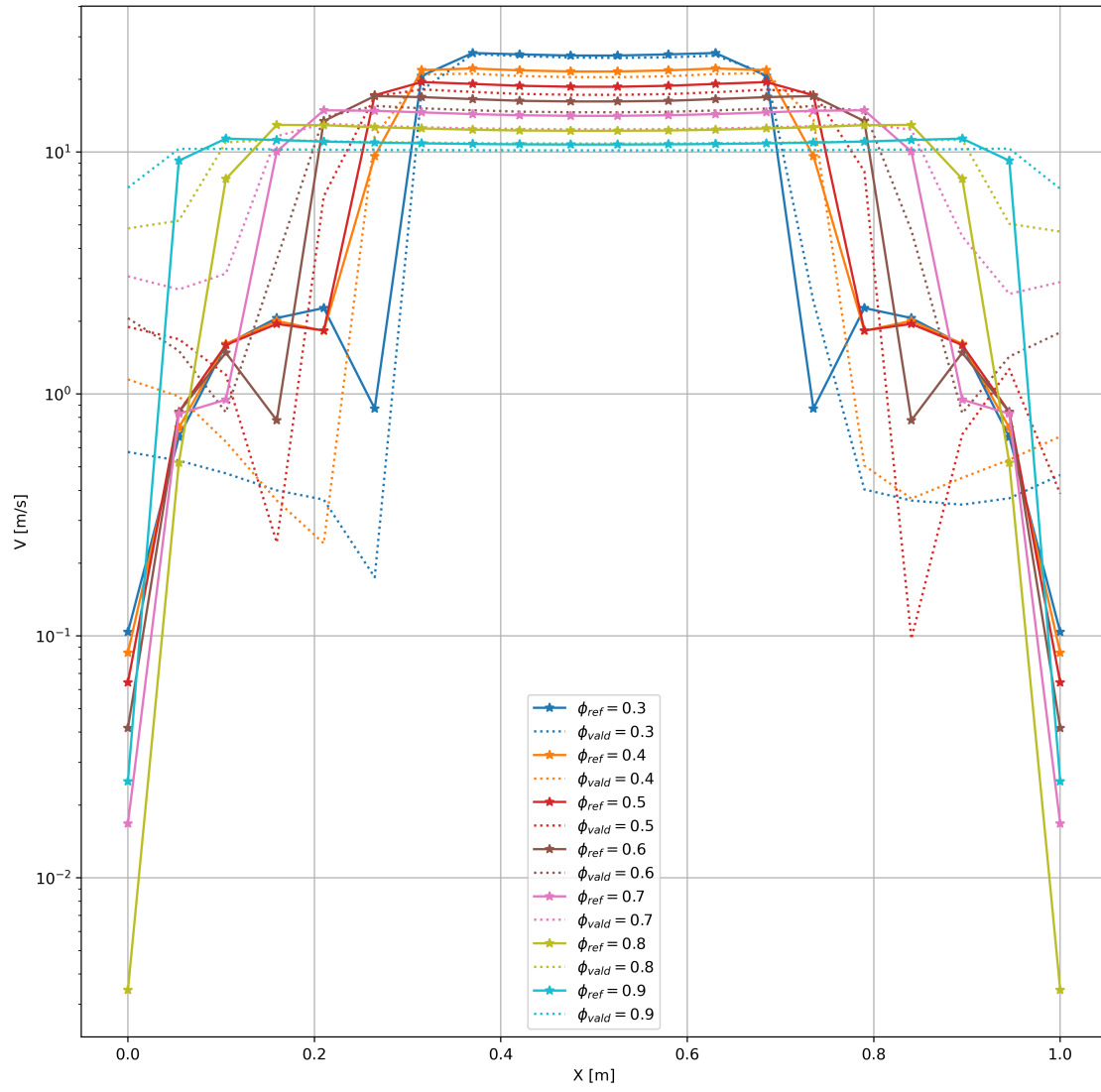


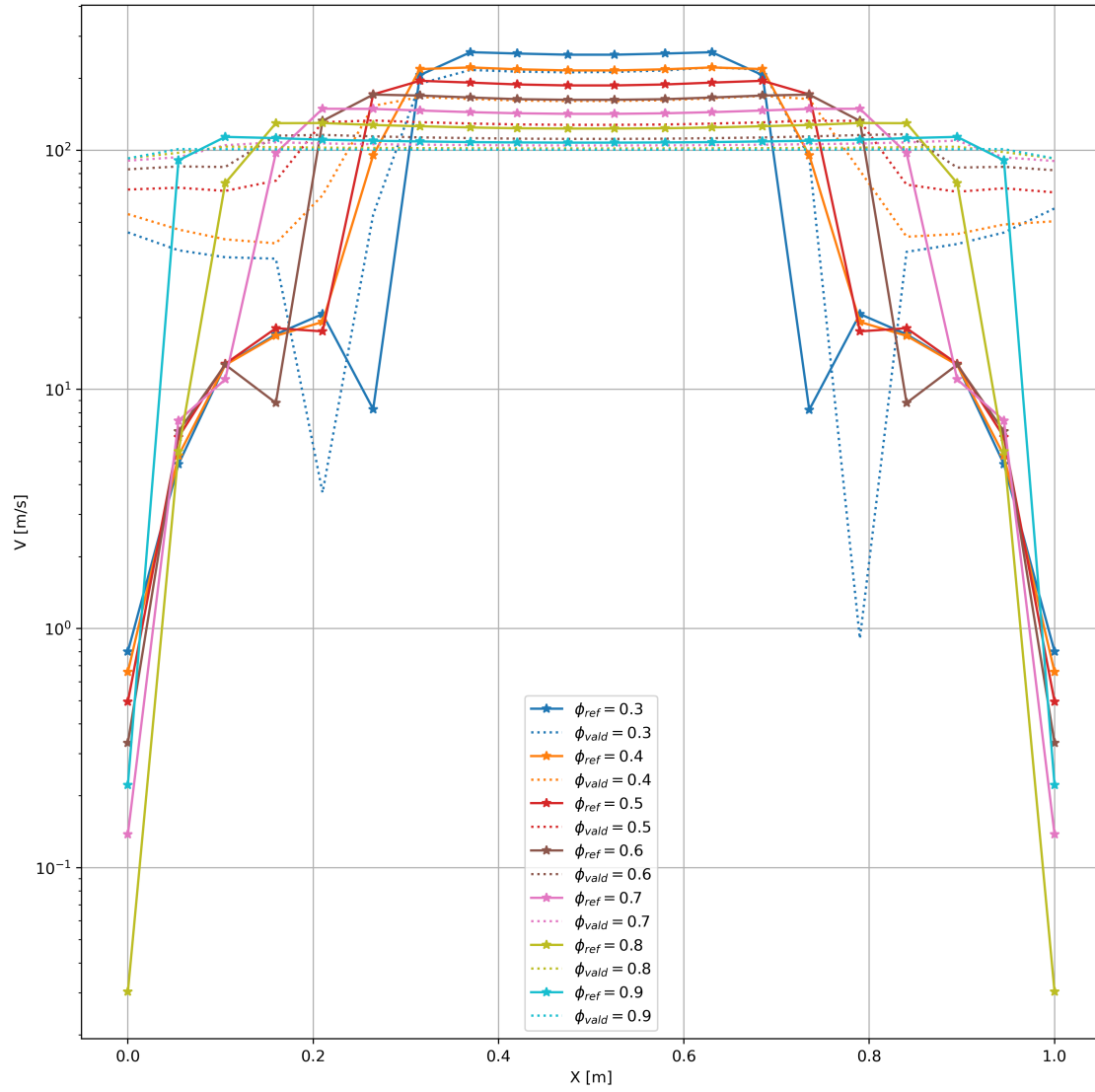


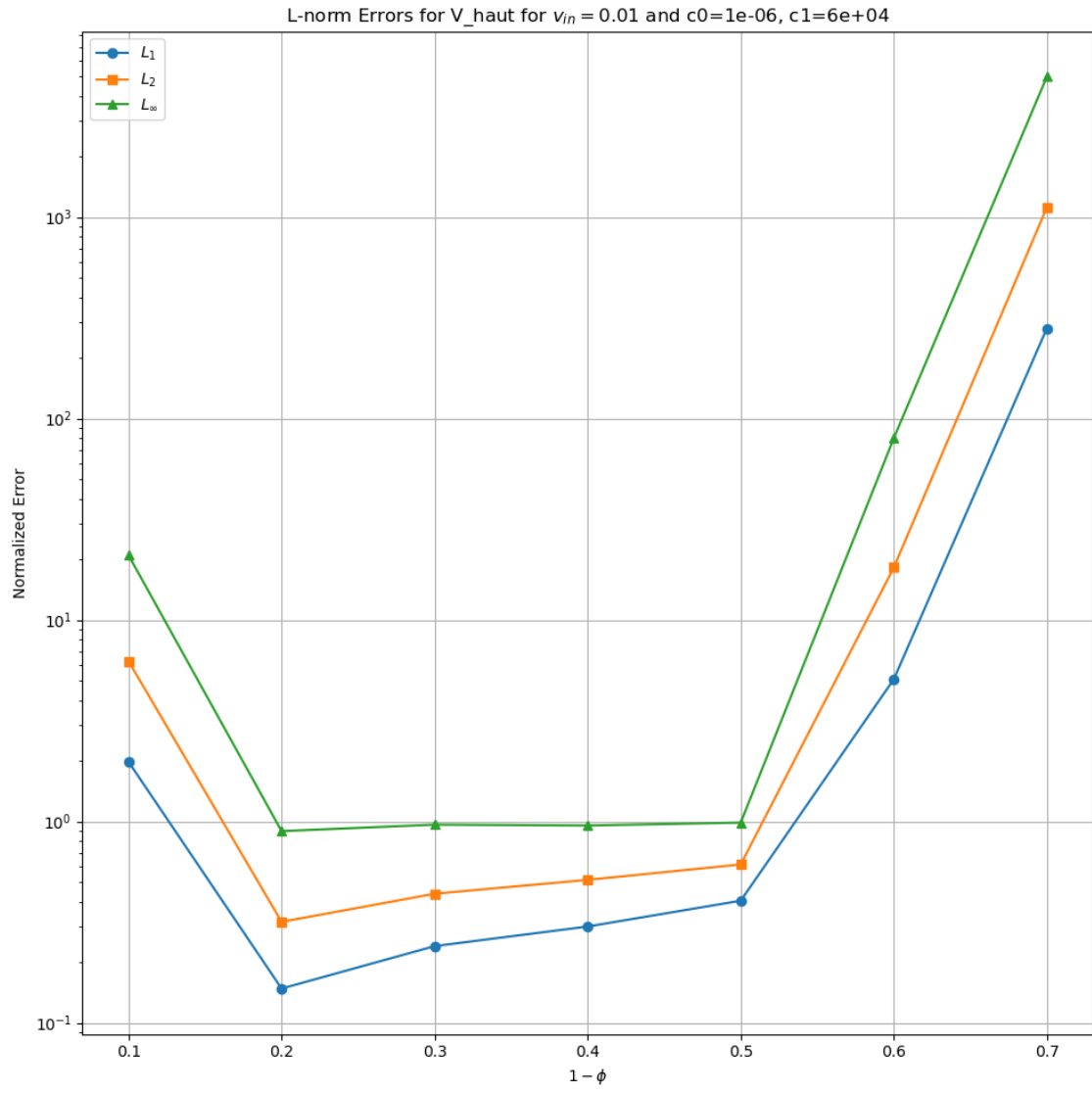


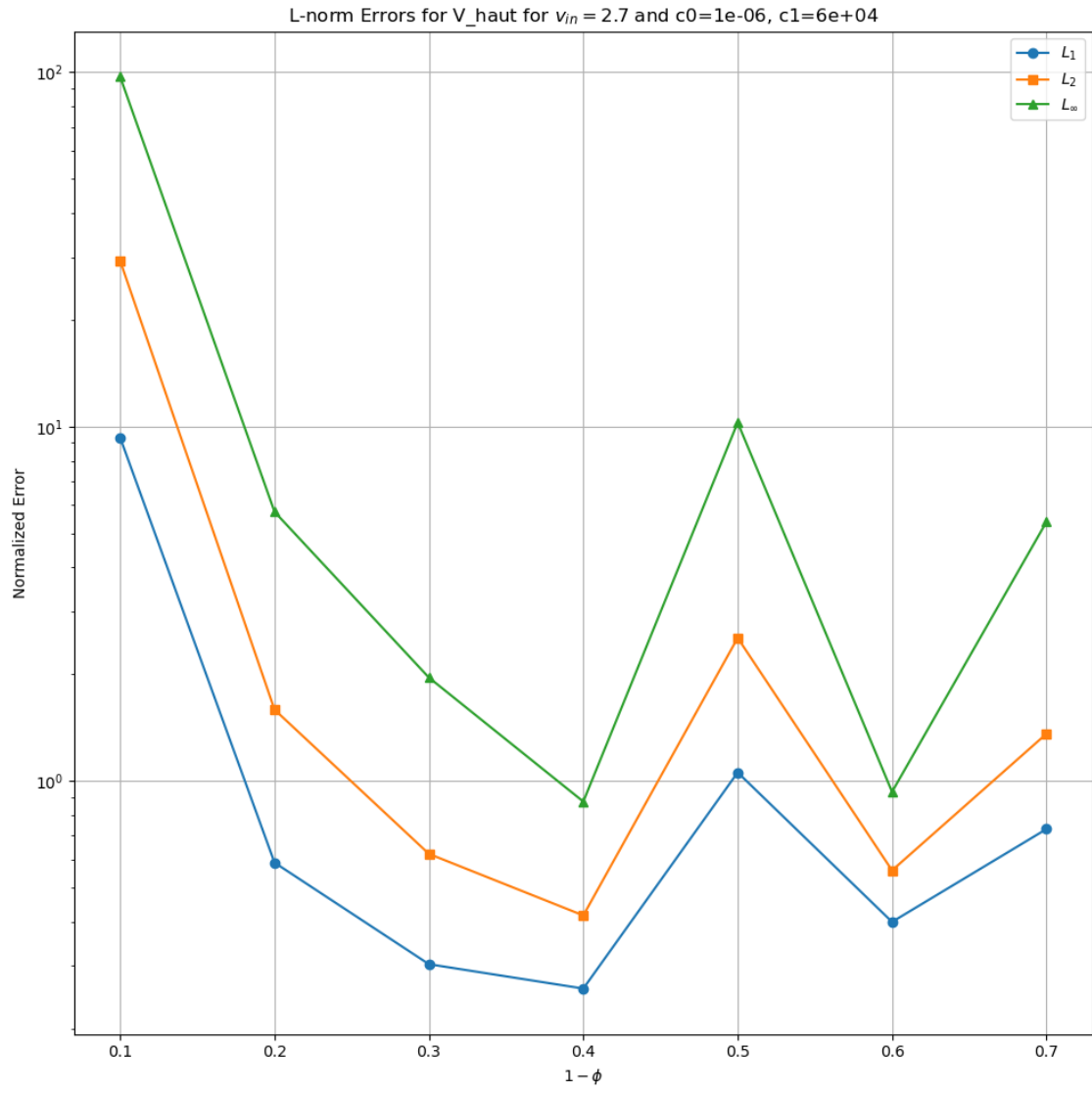


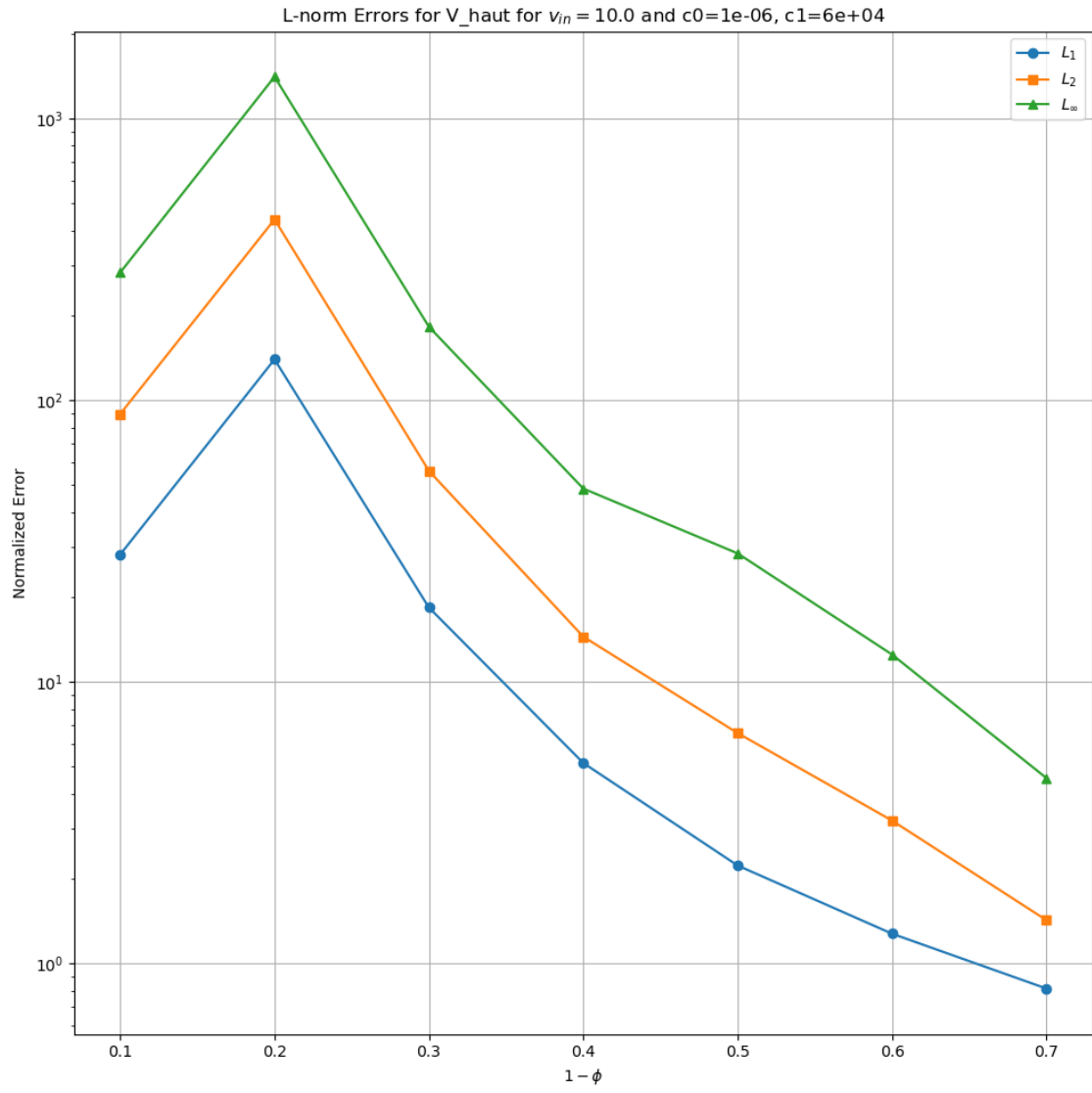


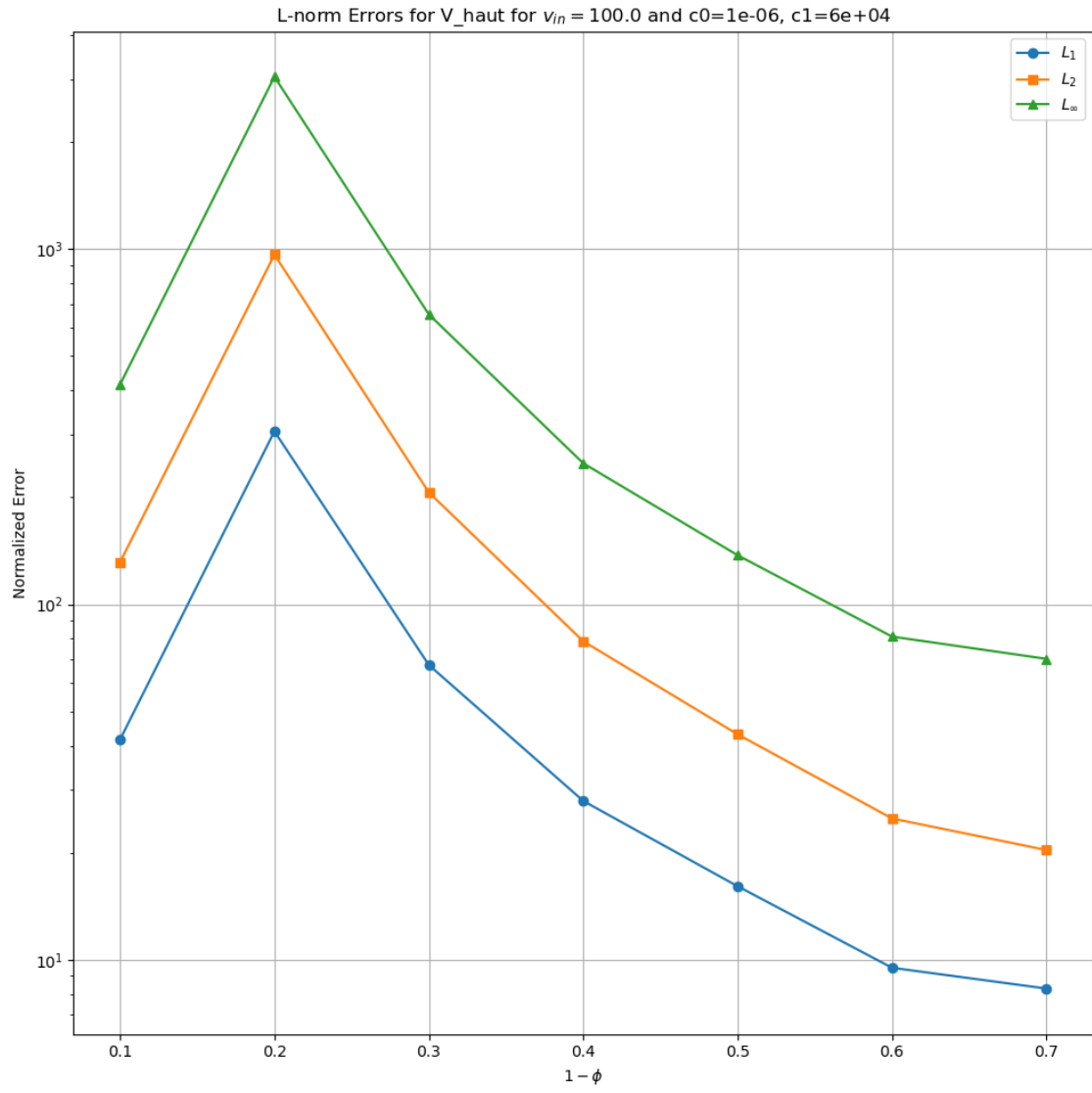


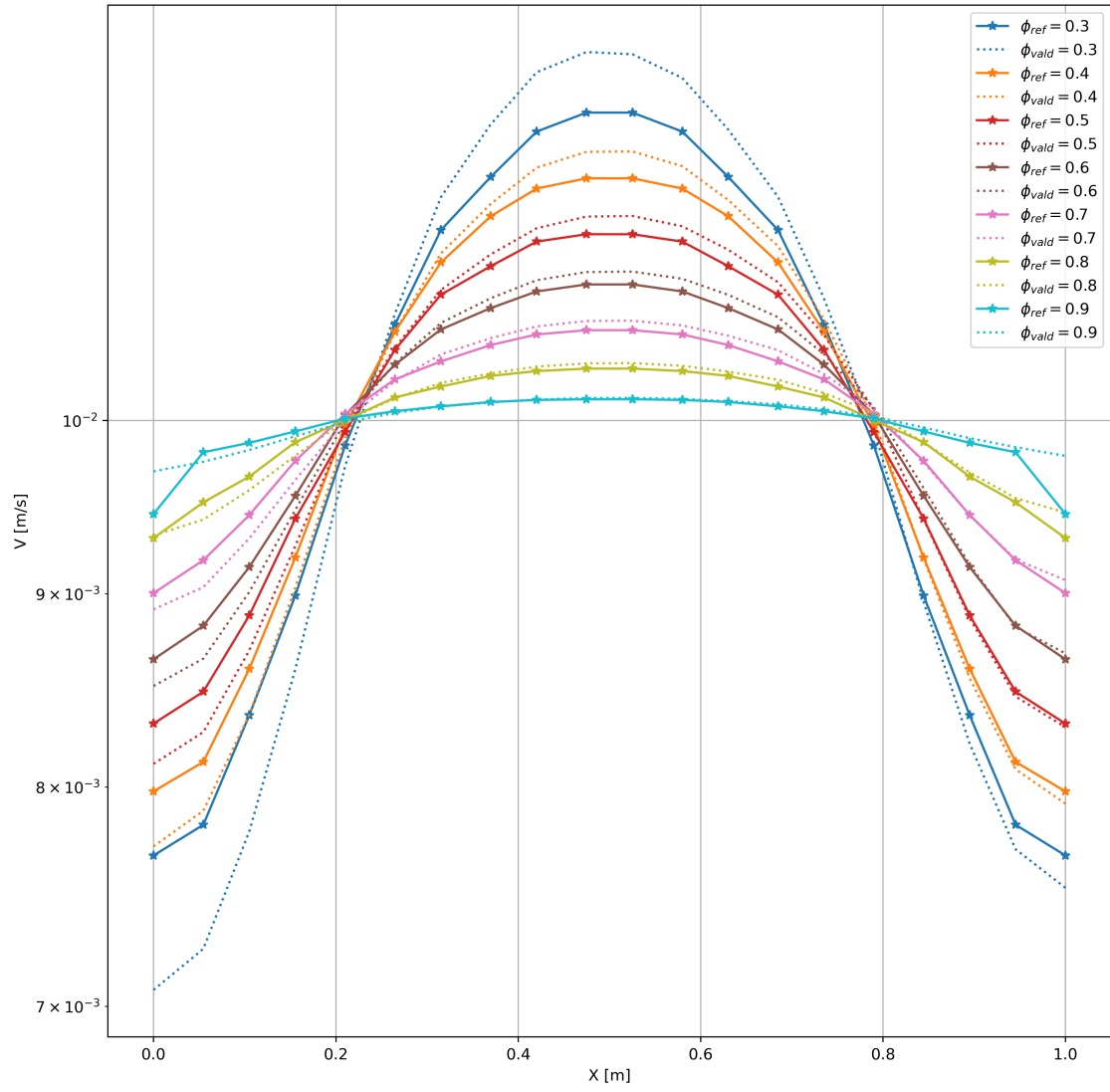


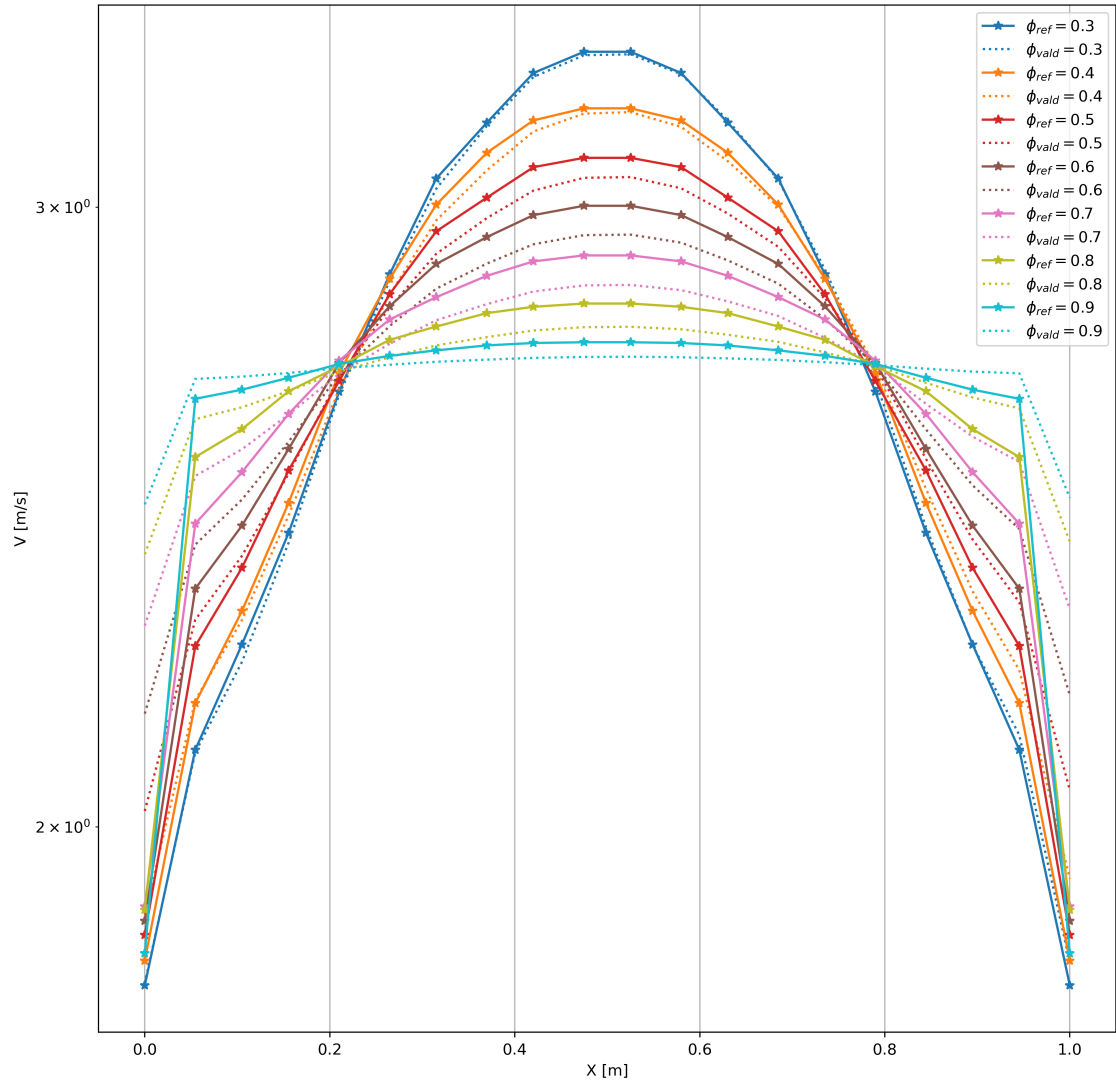


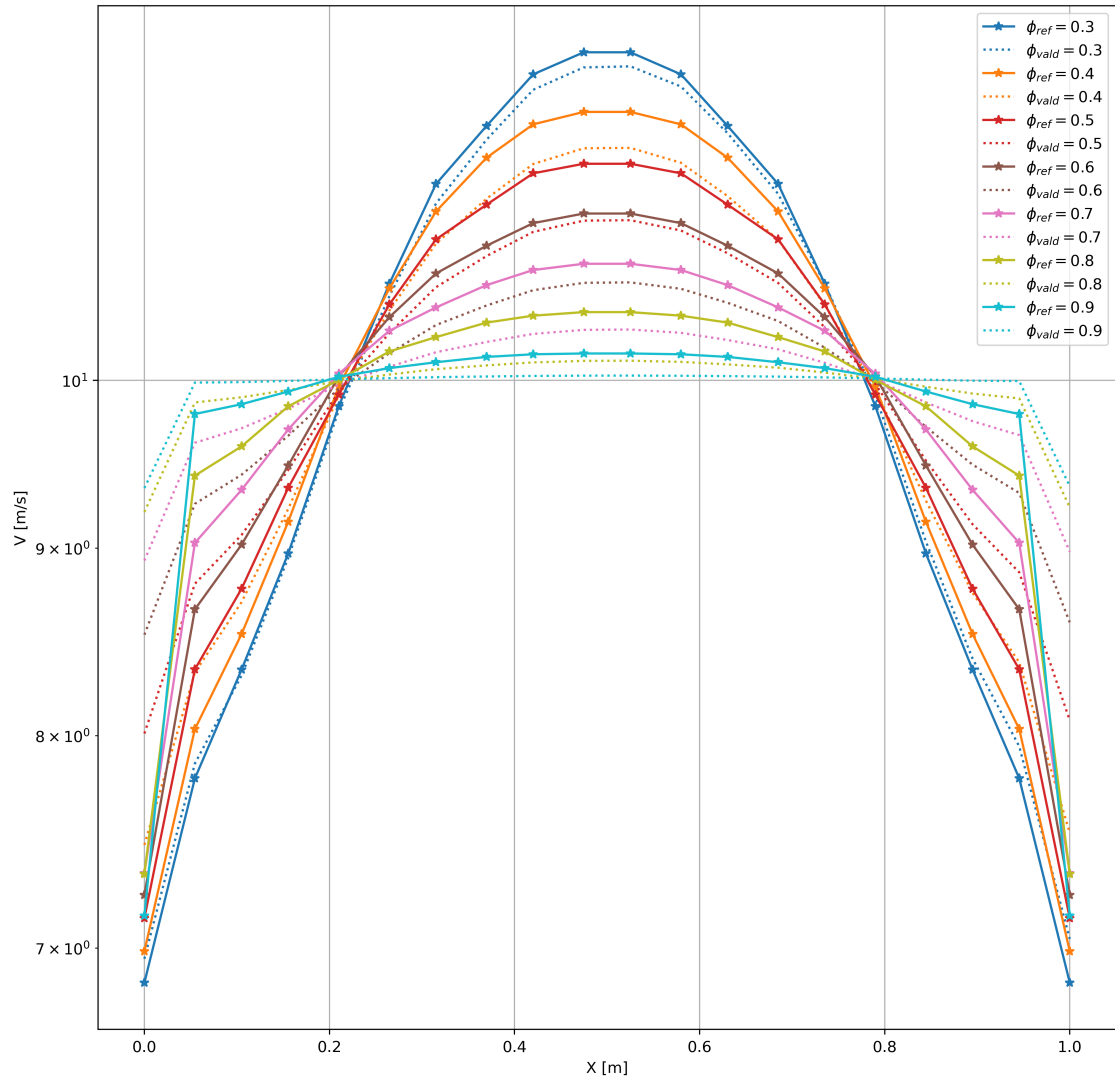


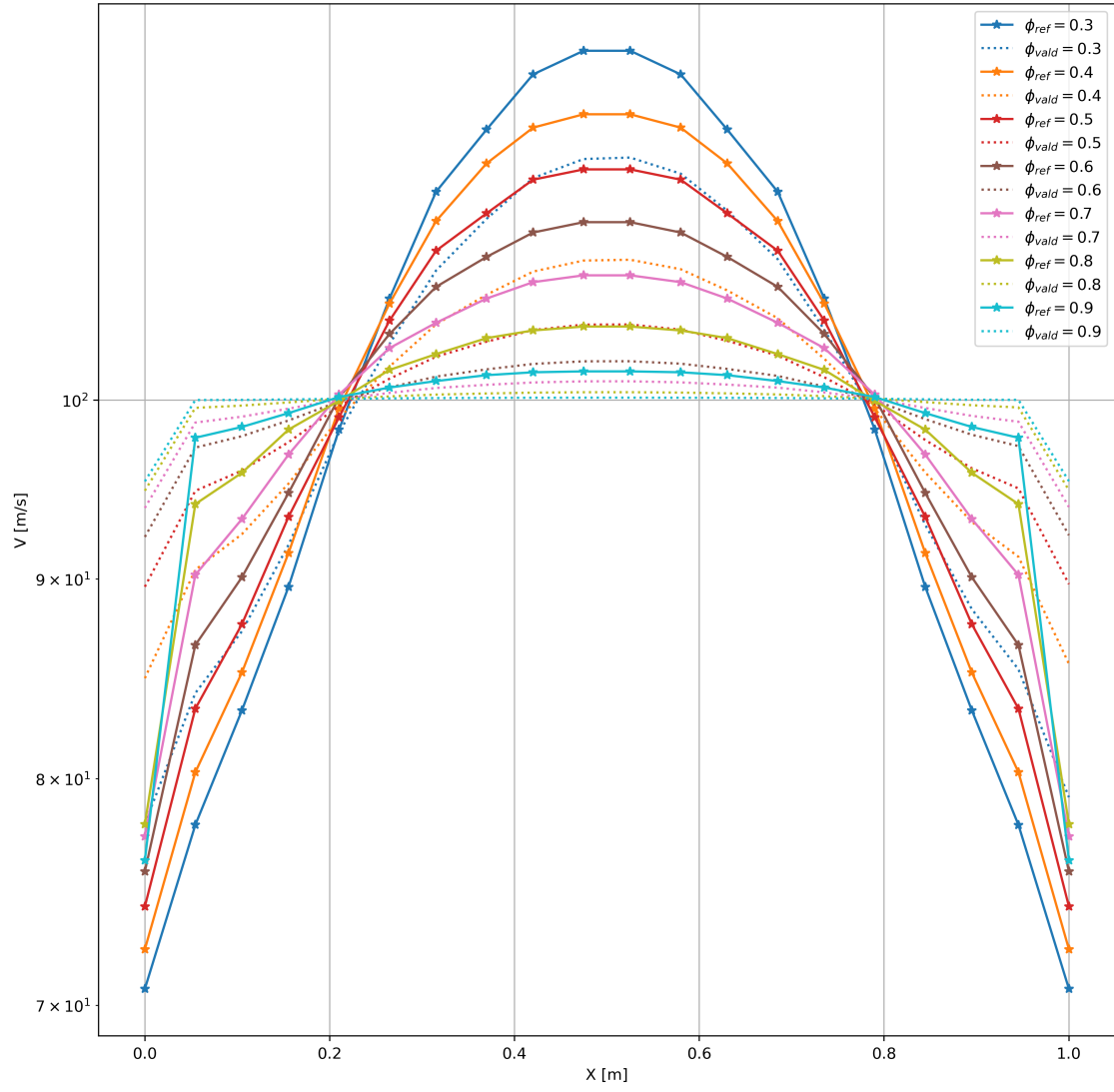


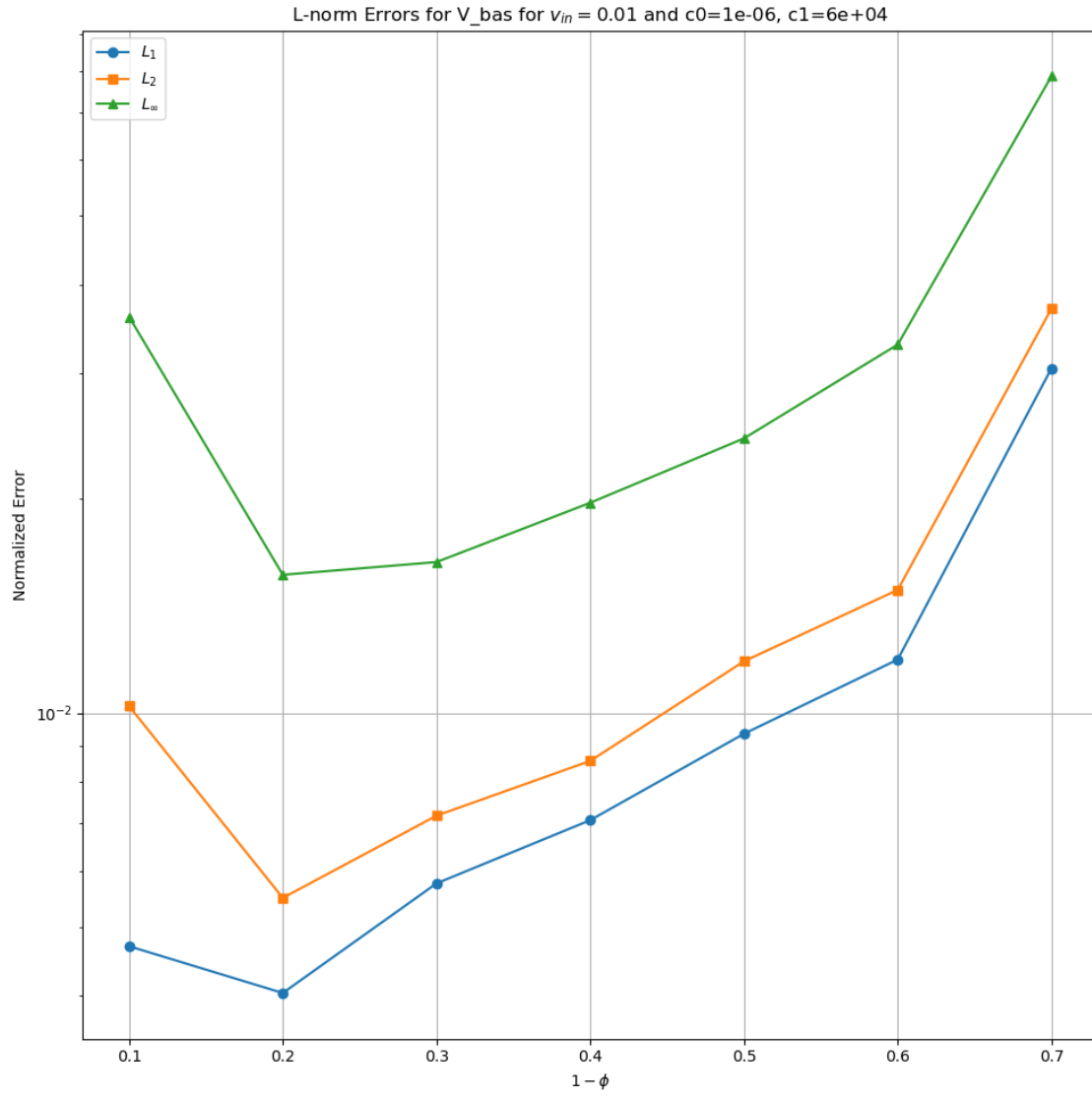


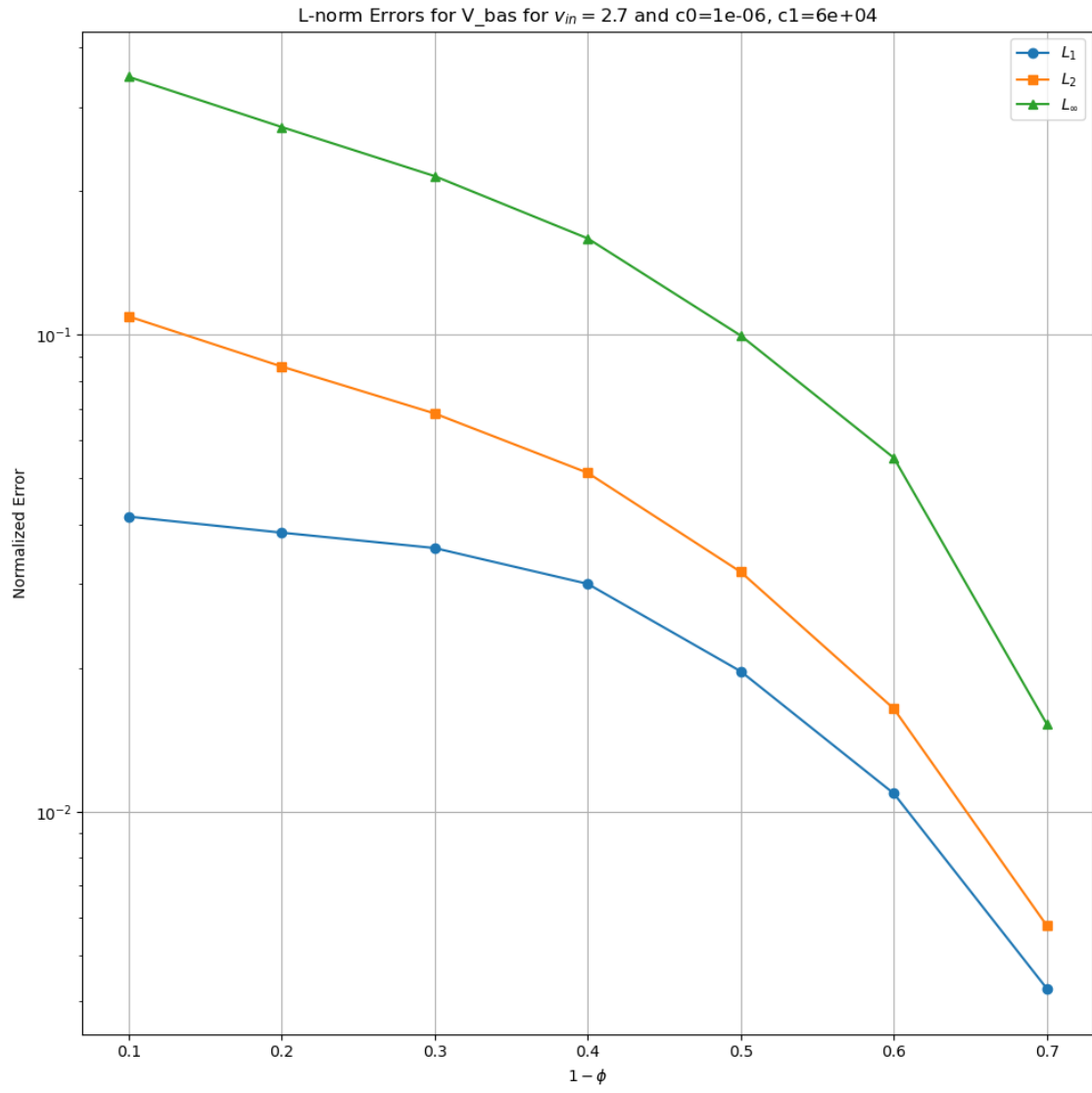


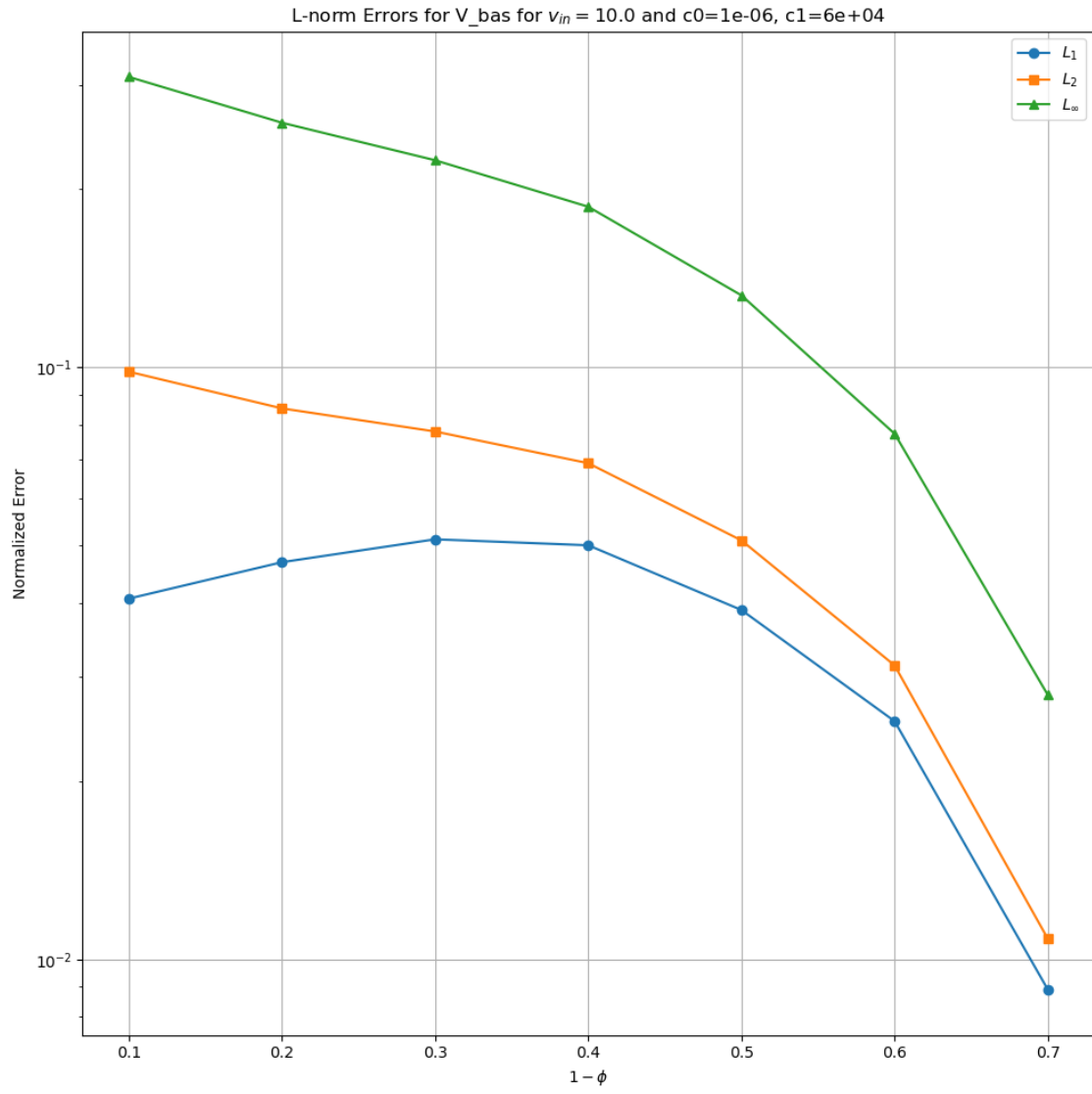


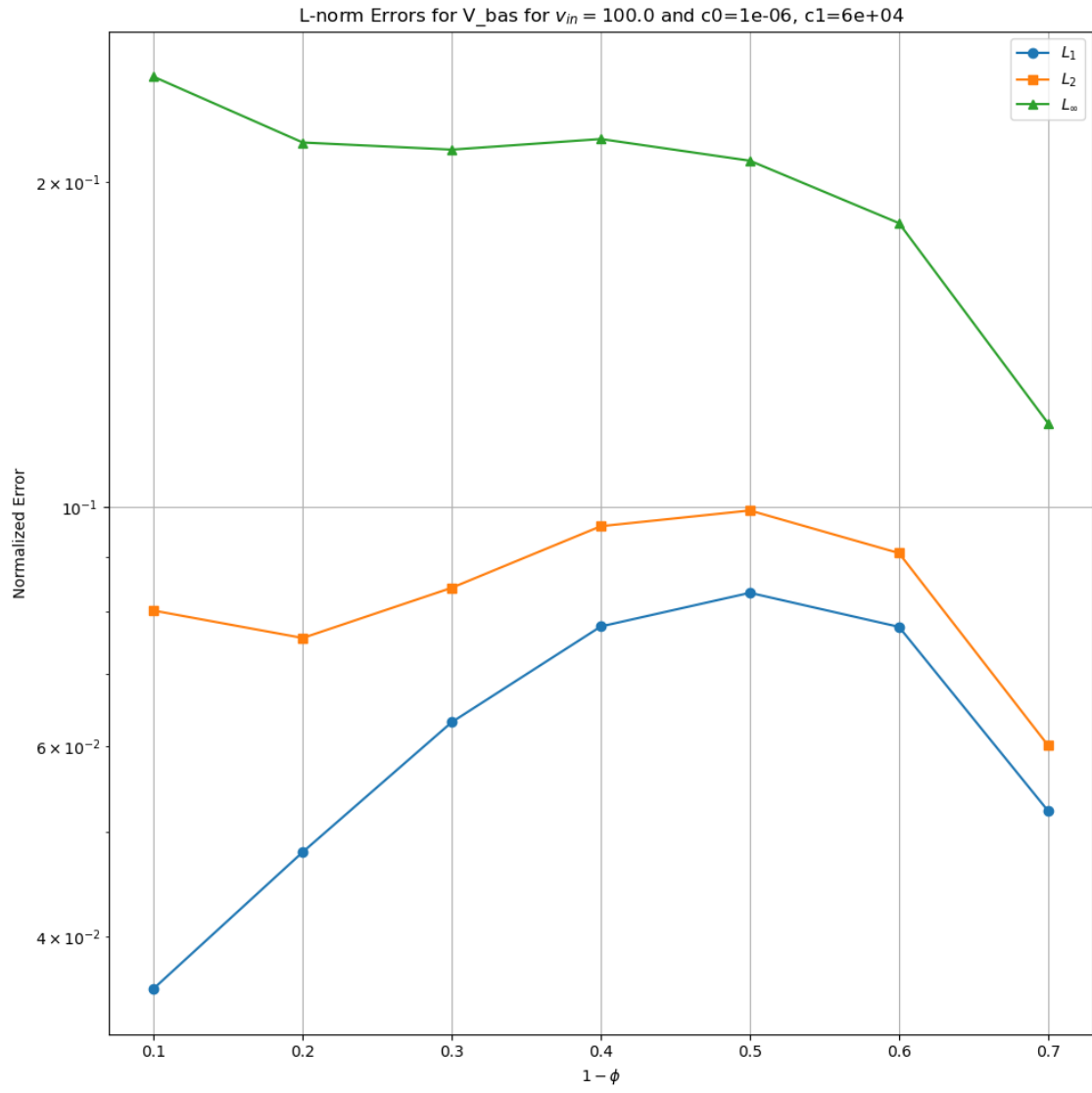


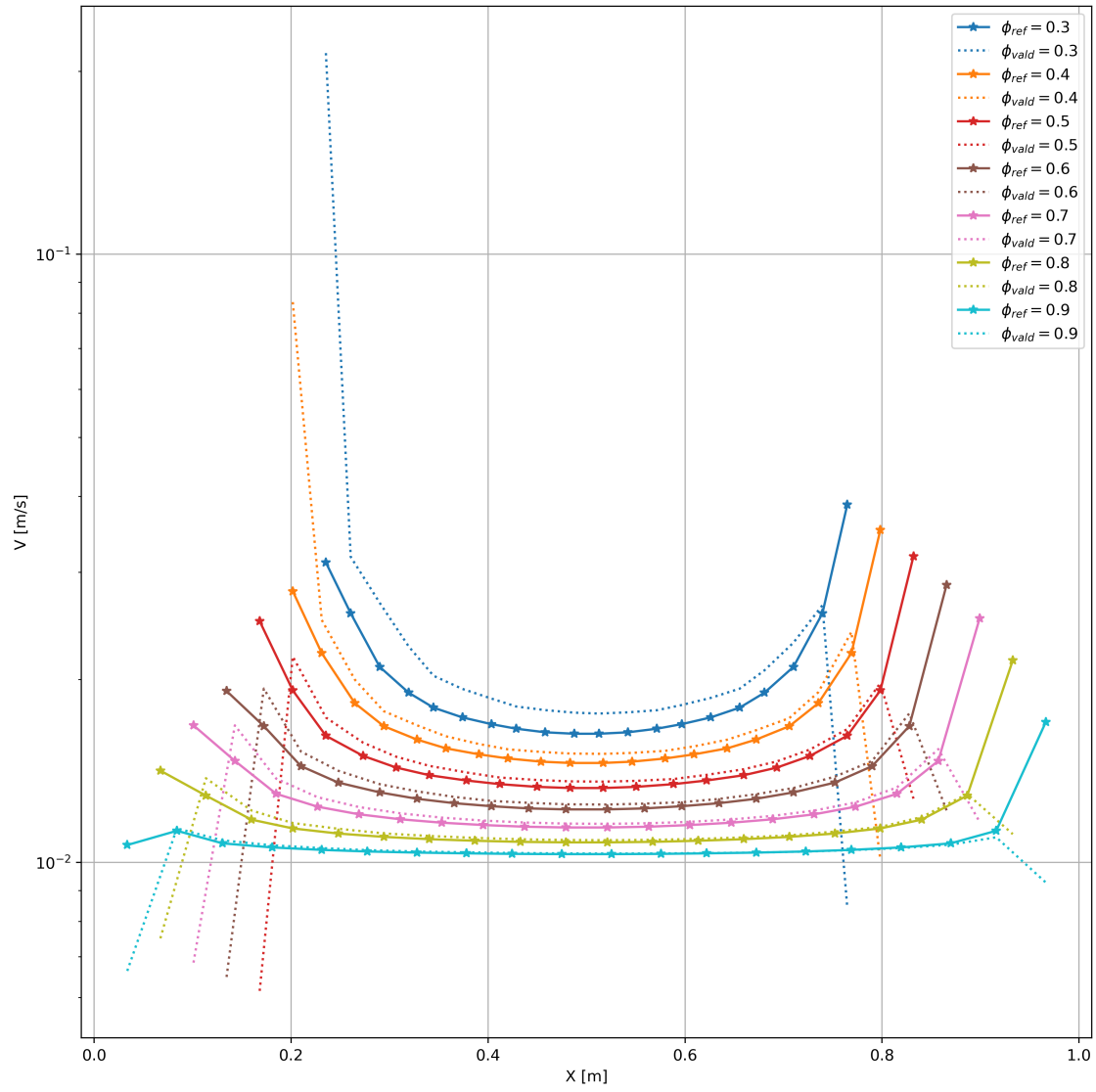


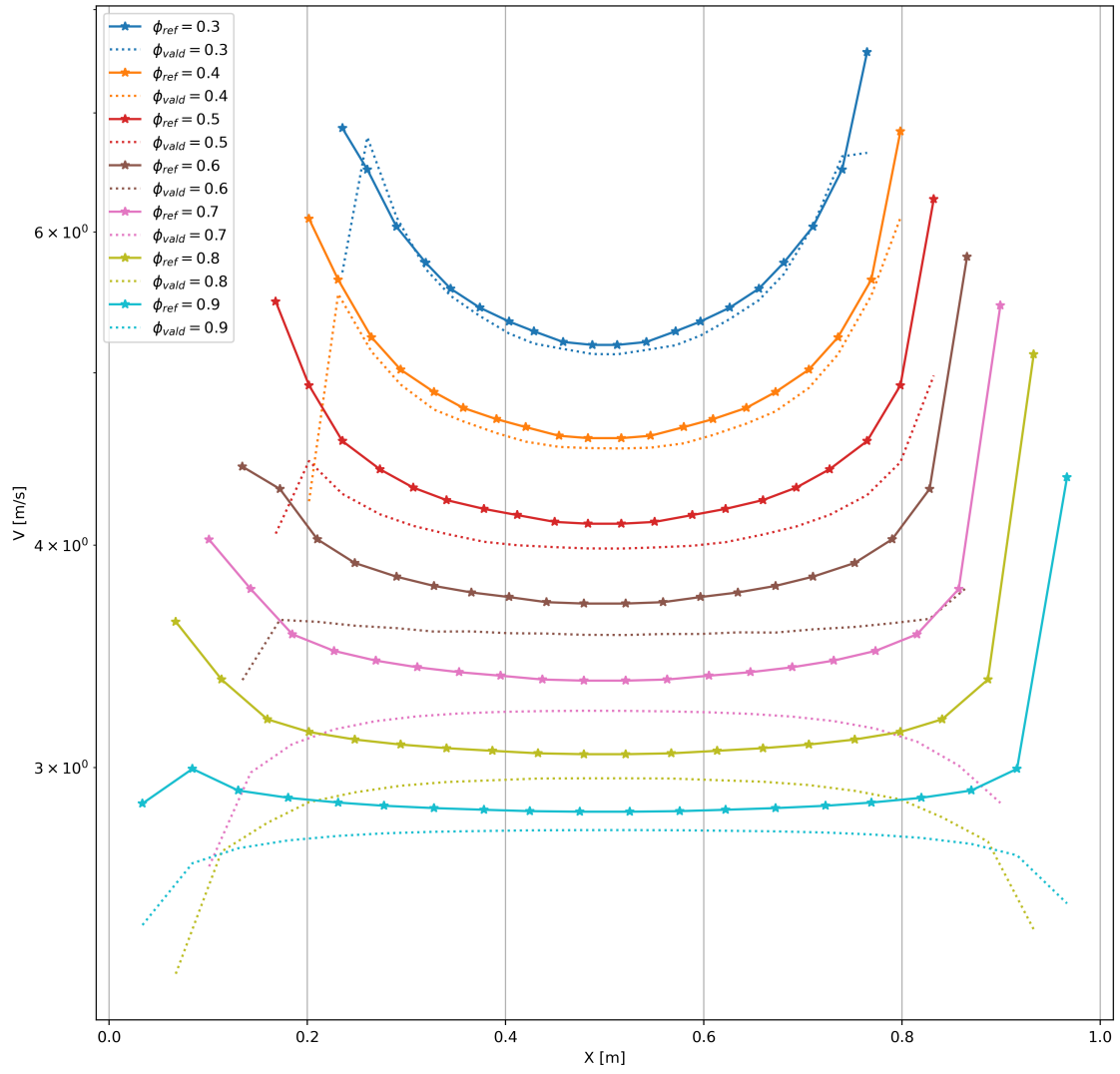


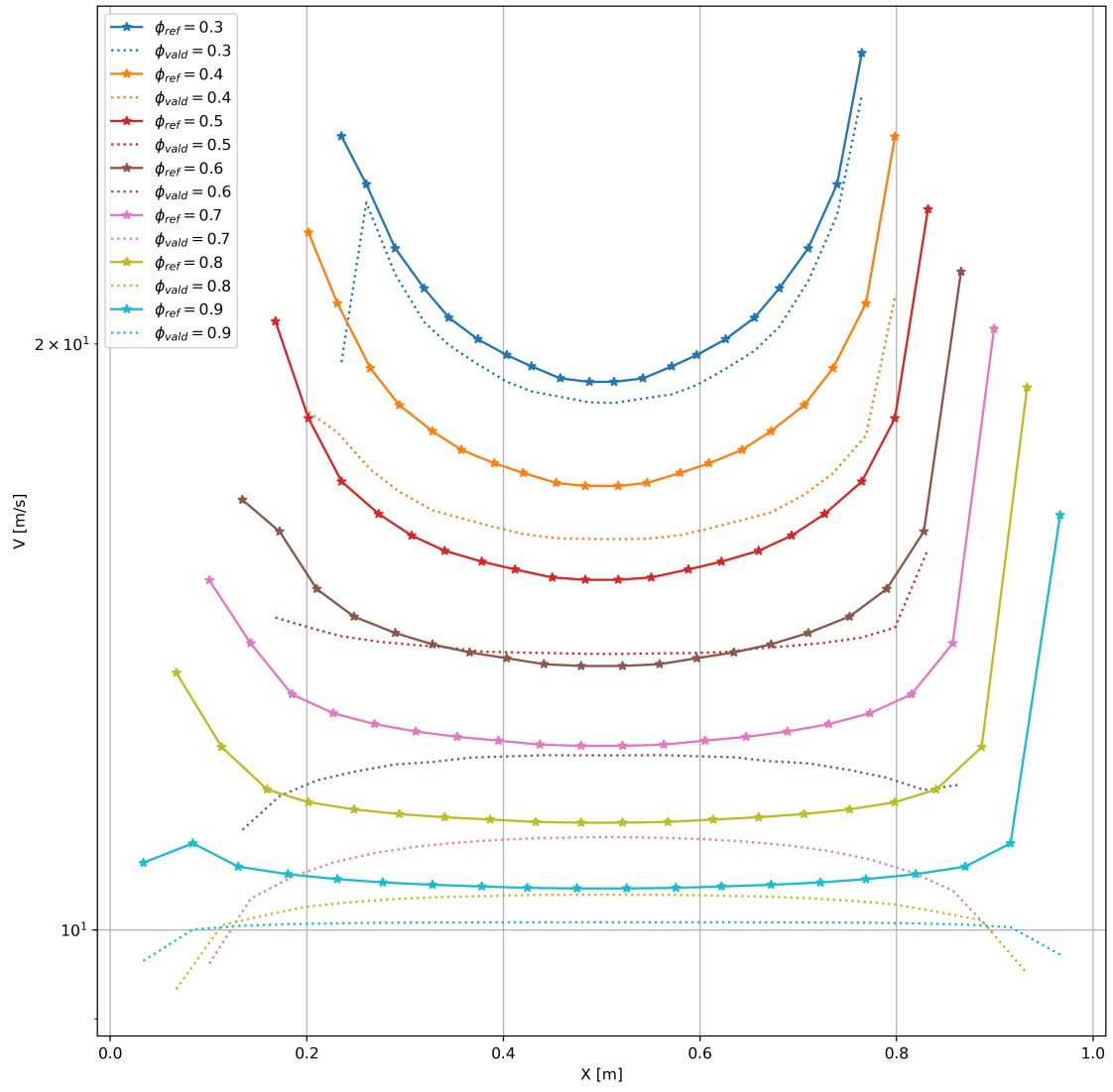


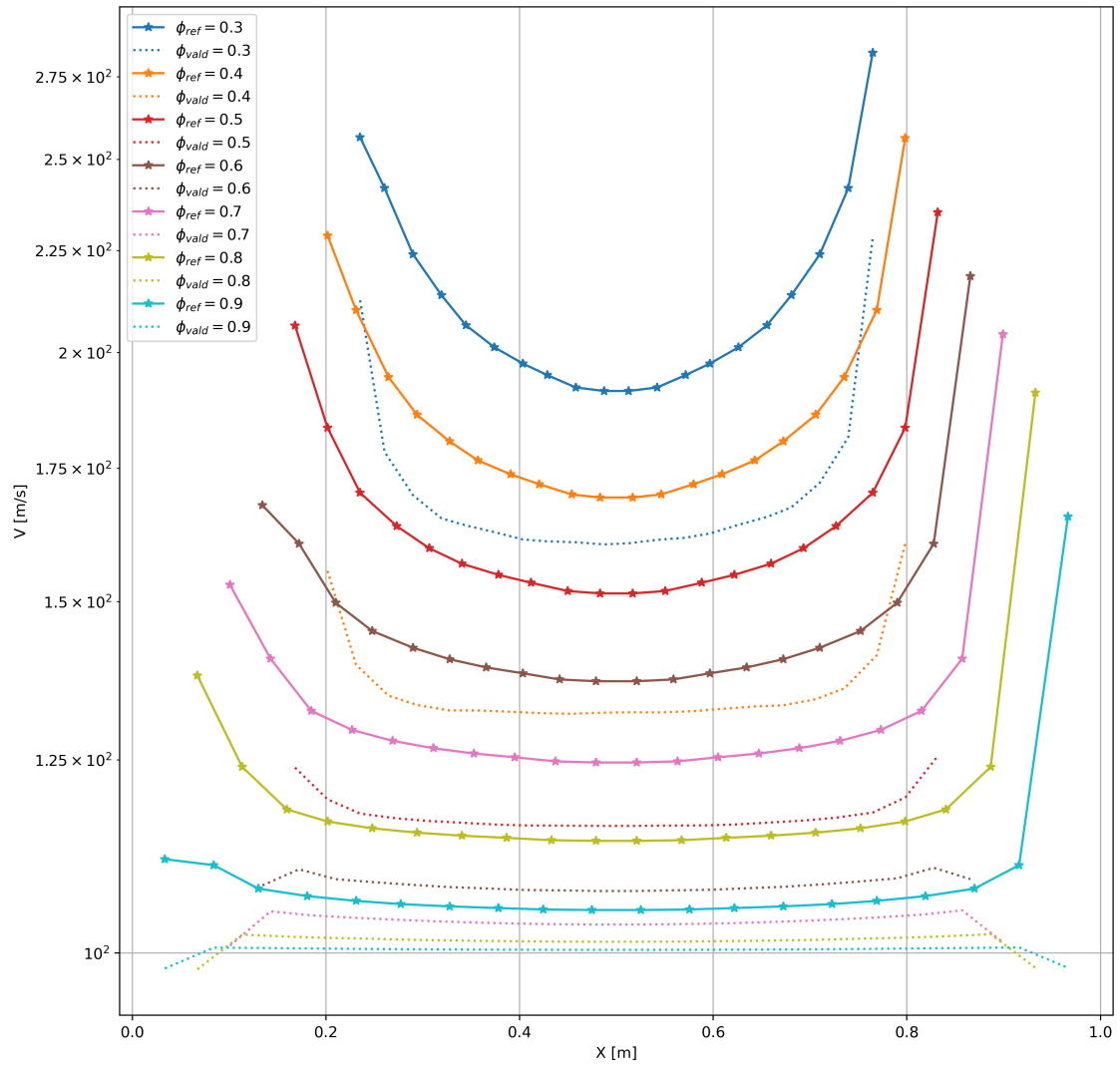


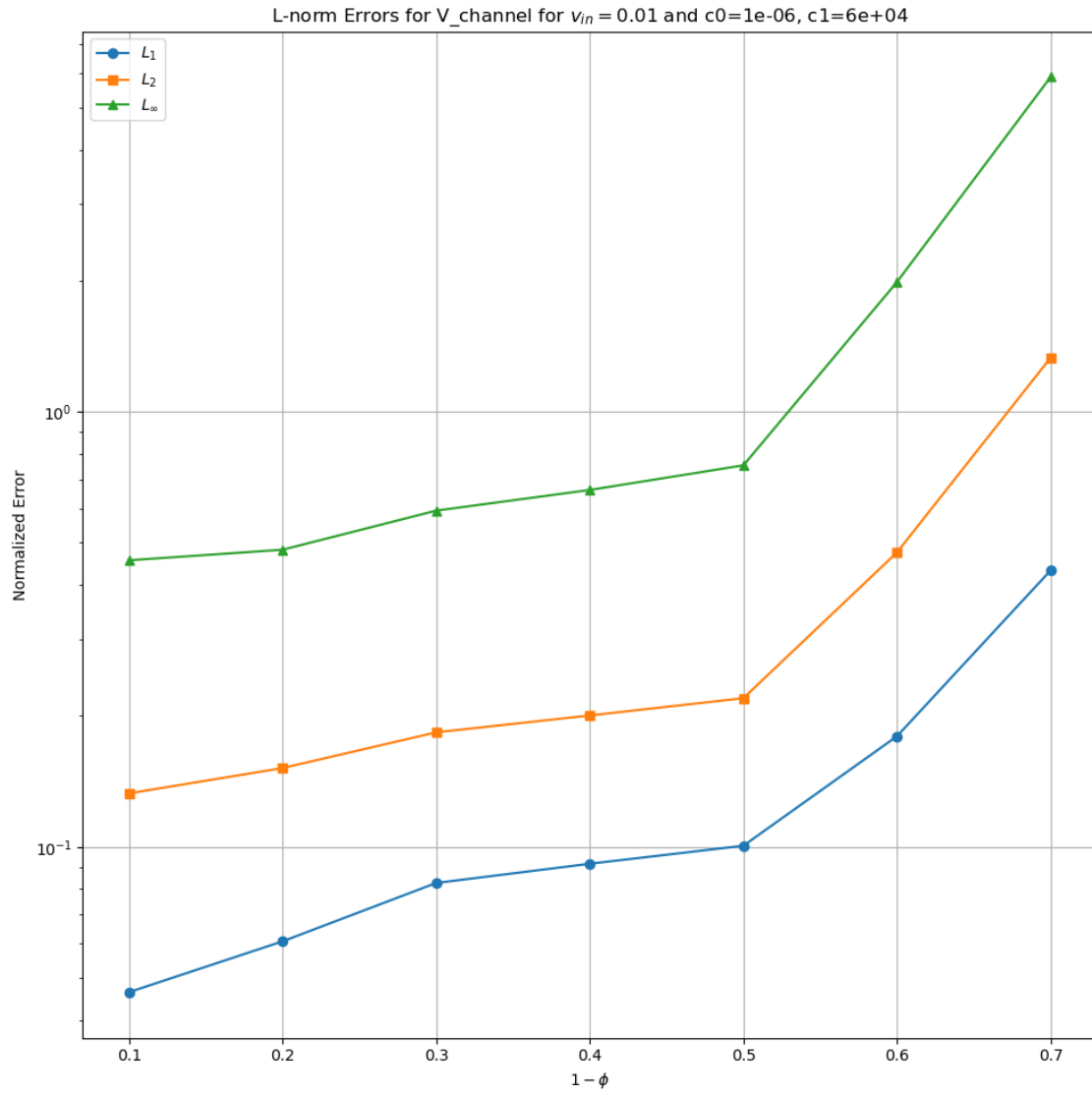


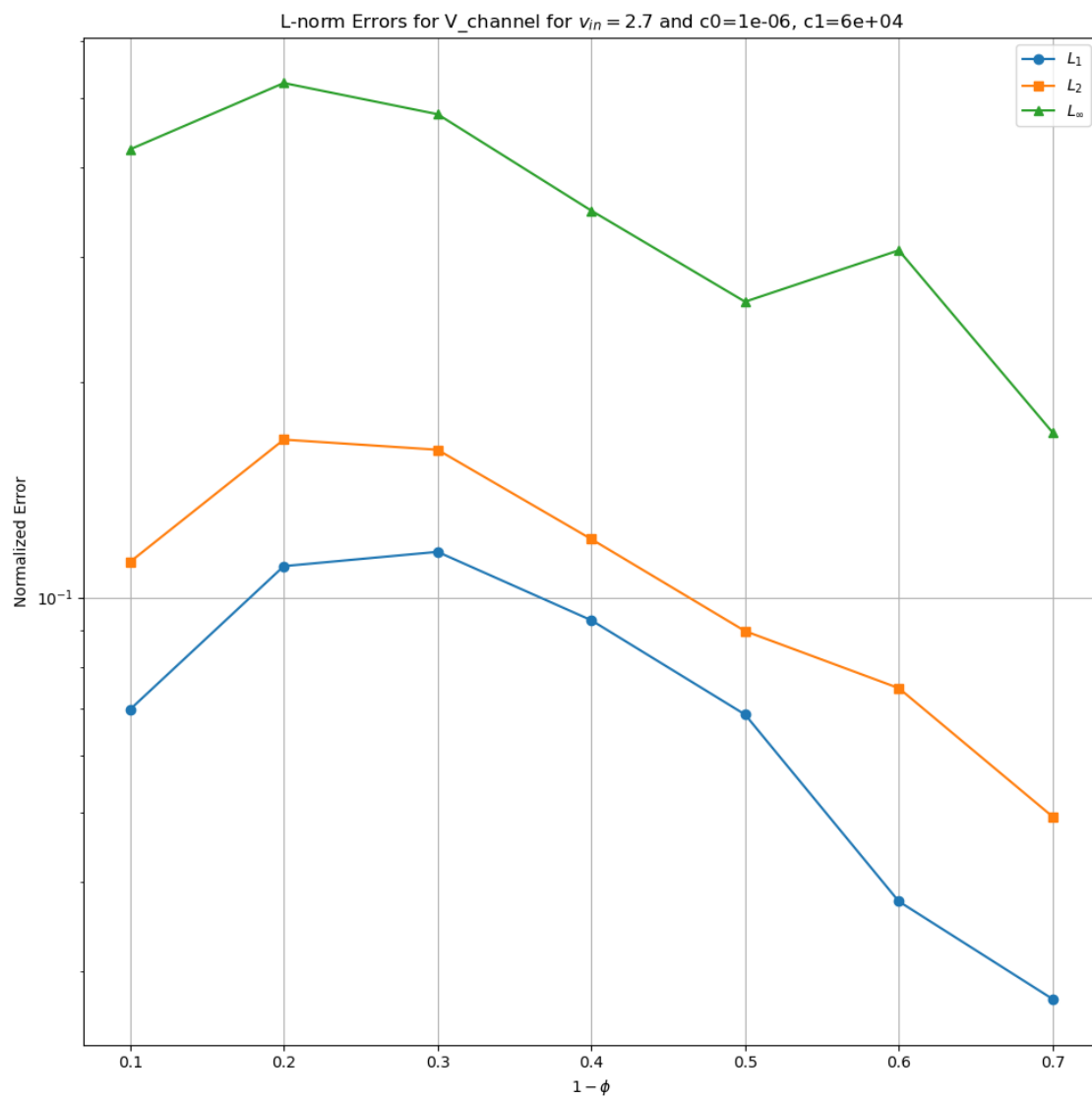


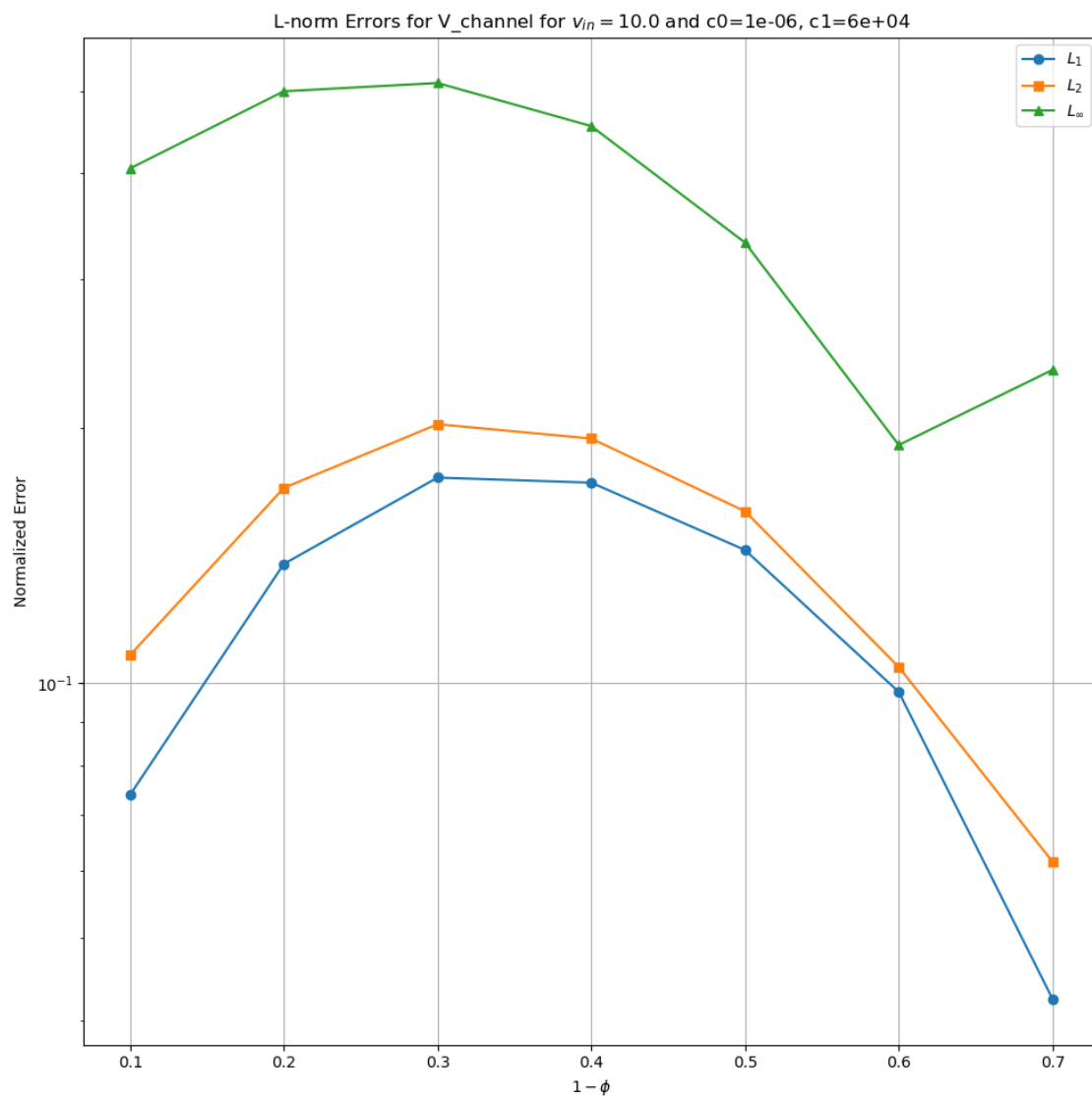


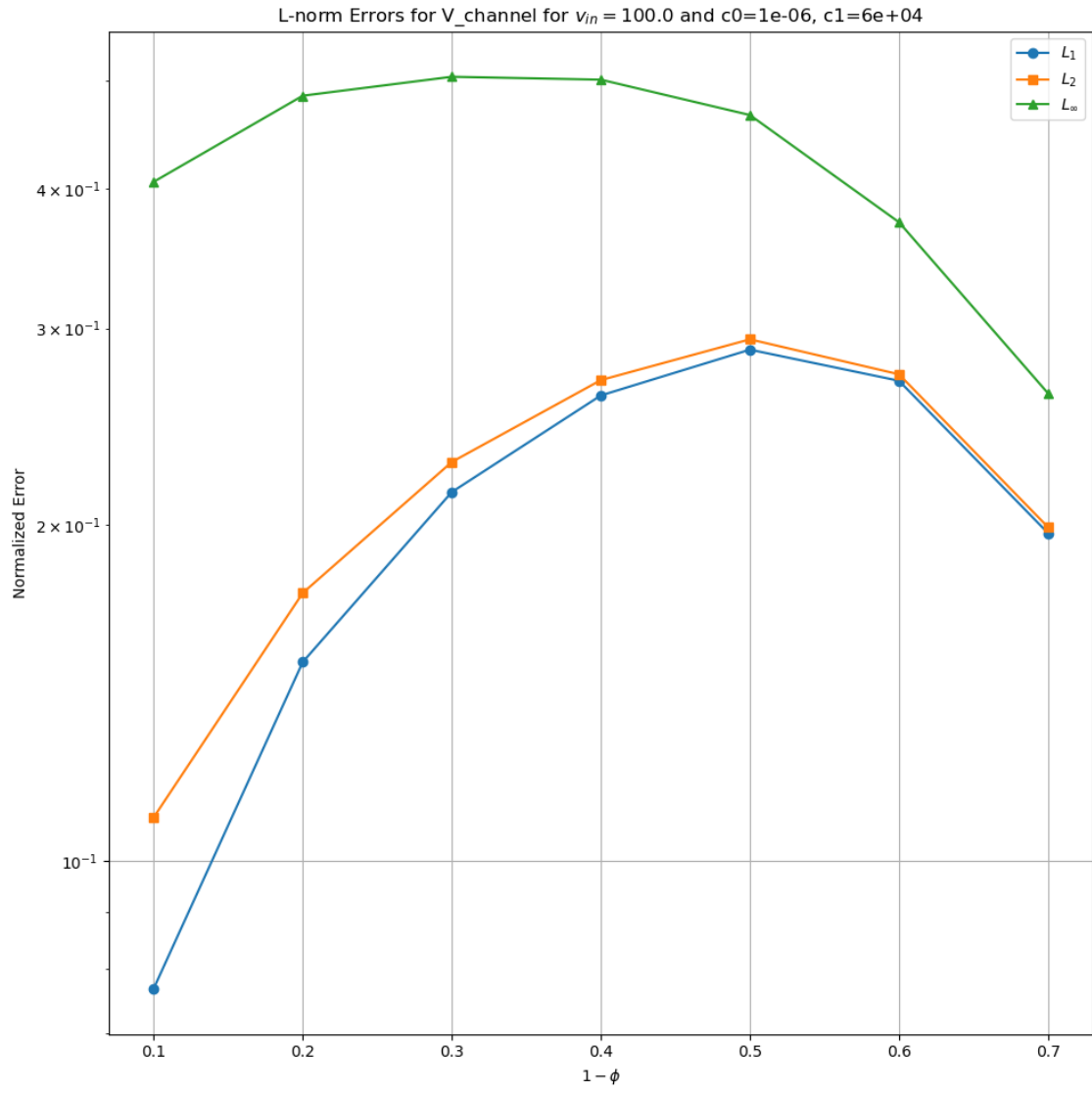


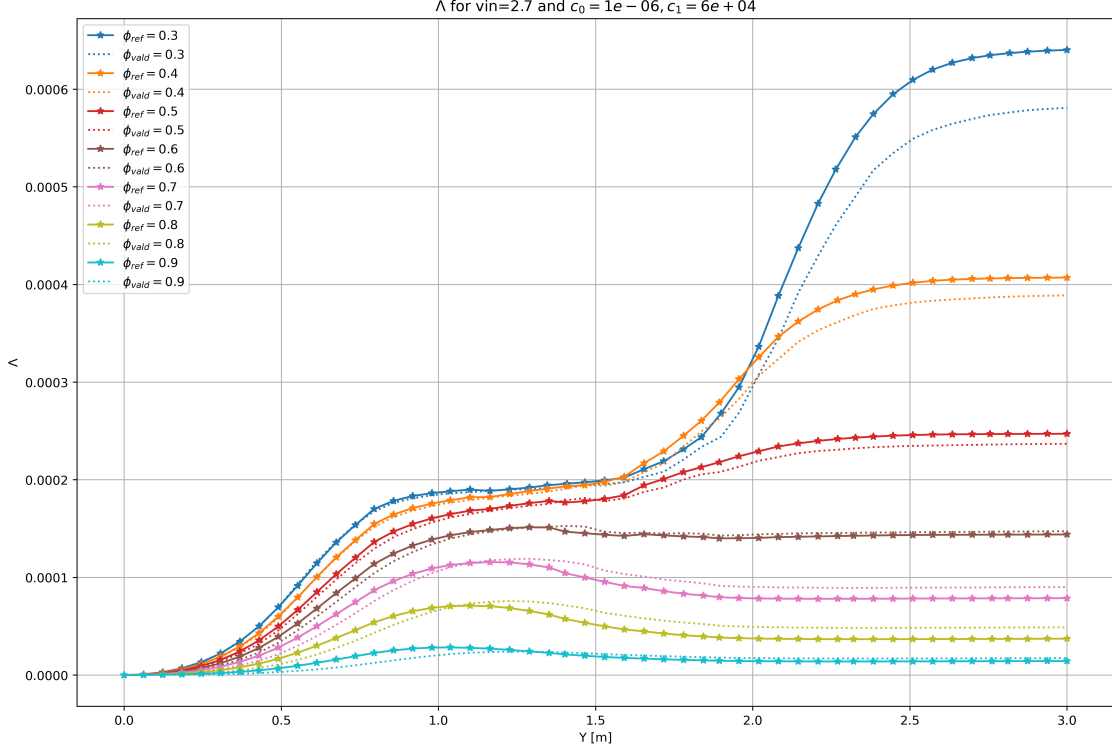












2 Error Evaluation

To evaluate the difference between the **validation model** and the **reference solution**, we compute dimensionless error metrics based on normalized L-norms.

2.1 Normalized L_1 Norm (Mean Relative Error)

This metric quantifies the **average** relative error across all data points.

$$L_1 = \frac{1}{n} \sum_{i=1}^n \left| \frac{\text{val}_i - \text{ref}_i}{\text{ref}_i} \right|$$

2.2 Normalized L_2 Norm (Root Mean Square Relative Error)

This norm emphasizes **larger deviations**, i.e. outliers.

$$L_2 = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{\text{val}_i - \text{ref}_i}{\text{ref}_i} \right)^2}$$

2.3 Normalized L_∞ Norm (Maximum Relative Error)

This gives the **worst-case relative error**, i.e. largest deviation.

$$L_\infty = \max_i \left| \frac{\text{val}_i - \text{ref}_i}{\text{ref}_i} \right|$$